GENETIC ASSESSMENT OF BOARDMAN RIVER FISH POPULATIONS BEFORE DAM REMOVAL

By

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ABSTRACT

The genetic assessment of fish populations in the Boardman River, MI presented here incorporated two unique approaches to address a variety of questions about population structure and species richness in this system. These two research projects are presented as two distinct chapters in this thesis, and each respective abstract is given below.

Fragmentation of river systems by dams can have a substantial genetic impact on fish populations. However, genetic structure can exist naturally at small scales through various processes such as isolation by adaptation (IBA) and spawning site fidelity, even in the absence of barriers. We sampled individuals from five native fish species with varying life histories above and below a dam in the lower Boardman River, Michigan, USA, and used RADseq to test whether genetic structure was influenced by the dam or other processes. Species assessed were white sucker *Catostomus commersonii*, yellow perch *Perca flavescens*, walleye *Sander vitreus*, smallmouth bass *Micropterus dolomieu*, and rock bass *Ambloplites rupestris*. We detected significant differentiation within each species, but this structure did not appear to be a result of fragmentation by the dam. Population groupings were not consistent with our original "above dam" and "below dam" sampled populations, but instead aligned with a Great Lakes (GL) group from Lake Michigan and a Boardman River (BR) group that appear to mix below the dam. We hypothesize that these groups formed prior to dam construction through IBA in these different habitats and further maintained divergence through spawning site fidelity. Additionally, GL fish for most species were significantly smaller in length than BR fish, suggesting a potential ontogenetic habitat shift of young GL fish into the lower river for feeding and/or refuge. Without our genetic assessment, the existence of these cryptic ecotypes likely would have continued undetected. Our study illuminates the importance of tributary habitats for GL fish and has major implications for the management of fish populations in the Great Lakes and

beyond. Finally, our approach of combining genetic data, ecological data, and simulations to assess connectivity and identify cryptic diversity has far reaching applicability for understanding the potential genetic impacts of fragmentation in other systems.

Understanding biodiversity in aquatic systems is critical to ecological research and conservation efforts, but accurately measuring species richness using traditional methods can be challenging. Environmental DNA (eDNA) metabarcoding, which uses high-throughput sequencing and universal primers to amplify DNA from multiple species present in an environmental sample, has shown great promise for augmenting results from traditional sampling to characterize fish communities in aquatic systems. Few studies, however, have compared exhaustive traditional sampling with eDNA metabarcoding of corresponding water samples at a small spatial scale. We intensively sampled Boardman Lake (137 ha) in Michigan, USA from May to June in 2019 using gill and fyke nets and paired each net set with lake water samples collected in triplicate. We analyzed water samples using eDNA metabarcoding with 12S and 16S fish-specific primers and compared estimates of fish diversity among methods. We captured a total of 12 fish species in our traditional gear and detected 40 taxa in the eDNA water samples, which included all the species observed in nets. The 12S and 16S assays detected a comparable number of taxa, but taxonomic resolution varied between the two genes. In our traditional gear, there was a clear difference in the species selectivity between the two net types, and there were several species commonly detected in the eDNA samples that were not captured in nets. Finally, we detected spatial heterogeneity in fish community composition across relatively small scales in Boardman Lake with eDNA metabarcoding, but not with traditional sampling. Our results demonstrated that eDNA metabarcoding was substantially more efficient than traditional gear for estimating community composition, highlighting the utility of eDNA metabarcoding for assessing species diversity and informing management and conservation.

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CHAPTER 1: GENETIC STRUCTURE IN FIVE FISH SPECIES FROM A FRAGMENTED RIVER IS DRIVEN PRIMARILY BY EXISTING HABITAT HETEROGENEITY RATHER THAN ISOLATION BY A DAM

INTRODUCTION

Contemporary genetic structure of wild populations is the result of multiple evolutionary and ecological processes that often act simultaneously on overlapping but variable timescales. Some of these processes include allopatric divergence resulting from isolation in different glacial refugia (Bailey & Smith, 1981; Bernatchez & Wilson, 1998; Sepulveda-Villet & Stepien, 2012), reproductive isolation resulting from spatial separation following recolonization (i.e. isolation by distance; Wright, 1943, 1946), population divergence as animals fill open habitats and adapt to different environmental conditions (i.e. isolation by adaptation/environment; Nosil et al., 2009; Orsini et al., 2013; Sexton et al., 2014), and genetic divergence resulting from fragmentation and recent barriers to gene flow (Keyghobadi, 2007). Disentangling the relative influence of these processes on contemporary genetic structure can be complex but is important for informing management and conservation action (Epps & Keyghobadi, 2015; Palsboll et al., 2007).

Dammed rivers are a well-known example of fragmented systems with evolutionarily recent barriers to gene flow. Some studies have observed clear effects of human-mediated fragmentation on the genetic structure of fish populations (Horreo et al., 2011; Raeymaekers et al., 2008; Yamamoto et al., 2004), while others have not (Clemento et al., 2008; Ruzich et al., 2019). For most systems, it can be difficult to disentangle the genetic effects of contemporary fragmentation from historic evolutionary processes (Epps & Keyghobadi, 2015; Ewers & Didham, 2006). One potential approach that can enhance our understanding of the relative

impacts of natural versus anthropogenic forces on genetic structure in fragmented rivers is to examine multiple species with varying life histories (e.g. Ruzich et al., 2019 and Blanchet et al., 2010).

Here, we investigate the relative influence of habitat fragmentation caused by a dam on the genetic structure of five fish species in the lower Boardman River, a tributary to the Laurentian Great Lakes. The Boardman River drains a 740 km^2 watershed in the northwest of the lower peninsula of Michigan, USA (Figure 1). the lower reach of the river flows through Boardman Lake, a natural drowned river mouth lake, before emptying into Grand Traverse Bay, an inlet of northeastern Lake Michigan. Most of the Boardman River watershed is isolated from Lake Michigan by the Union Street Dam, an earthen dam constructed in 1867 and located just 1.5 km upstream from the river mouth (Kalish et al., 2018). The dam has effectively blocked all upstream movement of most fishes with the exception of introduced Pacific salmonids which could ascend the pool and weir fishway while operational between 1986 to 2019 (Daniel Zielinski, GLFC, personal communication).

The Boardman River drainage is part of the Laurentian Great Lakes, a system characterized by large-scale habitat changes resulting from natural processes over the last several thousand years (Larson & Schaetzl, 2001). The Great Lakes were recolonized from multiple glacial refugia approximately 13,000 years ago after the Wisconsin glaciation, and broadscale genetic differences in fish populations exist among these glacial lineages (Borden et al., 2009; Sepulveda-Villet & Stepien, 2012; Stepien et al., 2009). Since recolonization, the Great Lakes have been highly dynamic (e.g. experiencing drastic water level fluctuations; Johnston et al., 2004), and fish have colonized diverse habitats. Intraspecific variation in life history strategies including fluvial versus adfluvial residence (Blumstein et al., 2018; Borden, 2008; Stepien et al.,

2007) and spawning philopatry (Leung & Magnan, 2011; Stepien et al., 2009; Strange & Stepien, 2007; Wilson et al., 2016) has also created sympatric divergence and fine scale genetic structure among adjacent or overlapping populations in the Laurentian Great Lakes. For example, Chorak et al. (2019) observed that yellow perch occupying drowned river mouths (DRMs) connected to Lake Michigan were genetically distinct from Lake Michigan perch, even though the Lake Michigan perch utilized DRM habitats during part of the year. Since the arrival of European settlers, humans have drastically altered the Great Lakes ecosystem through various activities (Regier et al., 1999; Ricciardi & MacIsaac, 2000), such as the construction of thousands of dams (Januchowski-Hartley et al., 2013). Contemporary genetic structure within Great Lakes fish species may thus be an artifact of multiple natural and anthropogenic processes that have occurred over both contemporary and evolutionary timescales.

We assessed the population genetic structure of five fish species that are native to the Laurentian Great Lakes and found both above and below the Union Street Dam: white sucker *Catostomus commersonii*, yellow perch *Perca flavescens*, walleye *Sander vitreus*, smallmouth bass *Micropterus dolomieu*, and rock bass *Ambloplites rupestris*. These species demonstrate varying life histories, generation times, and migratory behaviors (Becker, 1983; Scott & Crossman, 1985) (Table 1). For example, walleye and white sucker generally exhibit distinct migrations up tributaries in the spring to spawn (Becker, 1983) and tend to demonstrate natal homing (Crowe, 1962; Werner, 1979). Smallmouth bass, rock bass, and yellow perch may also migrate and exhibit spawning site philopatry, but movements and degree of homing tend to fluctuate among populations and systems (Brown et al., 2009; Gerber & Haynes, 1988; Glover et al., 2008; MacLean & Teleki, 1977). Substantial life history variations also exist within our study species. For example, smallmouth bass and rock bass are known to demonstrate both lake and

river life histories (Barthel et al., 2008; Gerber & Haynes, 1988; Noltie & Keenleyside, 1987), and these two ecotypes could remain genetically distinct even in the absence of geographic barriers (Borden, 2008; Euclide et al., 2020; Stepien et al., 2007). It is important to note that walleyes have been stocked extensively throughout the region including in Grand Traverse Bay and Boardman Lake (GLFC, 2020; MIDNR, 2020), and any natural genetic structure in the Boardman system may be masked by decades of stocking from non-native sources. To our knowledge, no evidence exists that any of our four other study species have been stocked in the Boardman system or in Grand Traverse Bay.

Our overall goal was to determine whether any observed population structure was the result of historic evolutionary processes or fragmentation by the Union Street Dam. Specifically, we (1) explored the population structure and diversity of our study species above and below the dam using genotypes from thousands of loci generated with restriction site-associated (RAD) sequencing, (2) conducted genetic migration simulations to compare our empirical data to simulated data generated under a variety of divergence scenarios, and (3) evaluated the relationship between observed patterns in population genetic structure and ecological attributes such as fish size and date of capture.

METHODS

Sample collection

We collected tissue samples from fish above and below the Union Street Dam for each of our five study species from 2017 to 2019. All sampling was performed with IACUC approval under University of Wisconsin – Stevens Point protocol number 2019.03.05. Fish below the dam were captured by boat electrofishing from the mouth of the Boardman River up to the Union

Street Dam. Field crews were unable to capture yellow perch in this river section, and we therefore acquired tissue samples from fish captured by ice anglers nearby (~ 1 km away) in Grand Traverse Bay. Above the dam, fish were collected in Boardman Lake using experimental gillnets, mini fyke nets, or boat electrofishing. Fish were identified to species, measured (mm total length), and tissue samples from the caudal or pelvic fins were collected and preserved in 95% ethanol. DNA was extracted from fin tissue with DNeasy® 96 Blood & Tissue Kits (Qiagen, Germantown, MD, USA).

RAD sequencing and SNP discovery

We prepared restriction site-associated DNA (RAD) libraries with the *SbfI* restriction enzyme following the BestRAD procedure (Ali et al., 2016) and methods outlined in Ackiss et al. (2020) with minor modification. Prepared libraries were sent to Novogene (Sacramento, CA) for sequencing on the Illumina NovaseqS4 platform (PE150 chemistry). The resulting sequences were processed through the STACKS v2.3 software pipeline (Catchen et al., 2011; Catchen et al., 2013) to demultiplex and filter raw reads, identify SNPs, and conduct genotyping. STACKS parameters for all species were as follows: *process_radtags* (-e SbfI, -c, -q, -filter_illumina, -r, --bestrad, -t 140), *ustacks* (--disable-gapped, --model_type bounded, --bound_high 0.05, -M 3, -max_locus_stacks 4, -m 3, -H, -p 32), *cstacks* (-n 3, -p 6 --disable_gapped). These STACKs parameters were derived based on review papers by Mastretta-Yanes et al. (2015) and Paris et al. (2017) and have been shown to work well for similar fish species with non-duplicated genomes (e.g. Bootsma et al., 2020). SNPs genotyped in > 30% of individuals (parameter flag: -r 0.3) were exported with the subprogram populations in variant call format (vcf) files. Filtering was then performed with vcftools v0.1.15 (Danecek et al., 2011) and included (1) removing loci genotyped in < 70% of individuals, (2) removing individuals genotyped at < 70% of loci, and (3)

removing loci with a minor allele count less than 3. We then used the program HDPlot (McKinney et al., 2017) to investigate read ratio deviation between alleles as well as locus specific heterozygosity and removed loci with heterozygosity greater than 0.60 or a read ratio deviation greater than 5 and less than –5. HDPlot is helpful for identifying potentially duplicated loci as well as identifying any potential laboratory or bioinformatic errors (e.g. contamination, inappropriate STACKs parameters) that may influence data quality. Finally, only the SNP with the highest minor allele frequency on each tag was included in the final dataset because loci on the same RAD tag may be linked.

Genetic differentiation and diversity

To assess and visualize genetic differentiation within each species, we first conducted principal component analysis (PCA) using the R package adegenet (Jombart, 2008). We also estimated the number of ancestral populations (*K*) contributing to contemporary structure for each species using the program ADMIXTURE v1.3 (Alexander et al., 2009). We tested *K* from 1-5 with ADMIXTURE's cross-validation procedure and then plotted *K*=2 through *K*=5 for each species in R. We then calculated pairwise F_{ST} and summary statistics for each population using the R package diveRsity (Keenan et al., 2013). Summary statistics included allelic richness, observed heterozygosity (H_o), expected heterozygosity (H_e), and inbreeding coefficients (F_{IS}). We tested the significance of each population comparison with a test for genetic differentiation (Goudet et al., 1996) conducted in GENEPOP ($\alpha = 0.01$). Using the R package related (Pew et al., 2015; Wang, 2011), we tested the relatedness of individuals within each species using the Wang pairwise relatedness estimator (Wang, 2002). Any pair of individuals with a relatedness value greater than 0.4 was considered highly related (likely siblings or parent/offspring).

Effective population size (N_e) of each population was estimated with the bias-corrected linkage disequilibrium method (Hill, 1981; Waples, 2006; Waples & Do, 2010) in the software package NeEstimator v2.1 (Do et al., 2014) with a p-crit of 0.05 (Waples et al., 2016). Physical linkage can bias estimates of N_e downward; therefore, we used genome resources to restrict comparisons to markers on different chromosomes when a genome assembly was available (yellow perch, walleye) and implemented a formula for correcting bias based on chromosome number for the other species (smallmouth bass, rock bass, white sucker; equation 1a in Waples et al., 2016). Alignments to the yellow perch genome for yellow perch and walleye, which shares the same karyotype (Danzmann, 1979), were conducted in BLASTN (Camacho et al., 2009); the best alignment for each locus was retained, and all alignments had e-values < 1 e^{-51}. Chromosome numbers for species where genomes were not available are as follows: N=23 for smallmouth bass (Beçak et al., 1971), N=24 for rock bass (Avise & Gold, 1977), and N=50 for white sucker (Beçak et al., 1973). N_e calculations using the linkage disequilibrium method can be biased slightly downward when individuals from multiple cohorts are included in the sample due to a slight Wahlund effect (7% downward bias on average; Waples et al., 2014), but this small bias should not greatly affect the interpretation of the N_e results.

Visualization of PCA and ADMIXTURE plots for most species suggested population groupings that were not consistent with our originally sampled "above dam" and "below dam" populations, but more broadly with what we hereafter refer to as a "Great Lakes" (GL) population and a "Boardman River" (BR) population (see Results for further explanation). To assess this unexpected structure, we reformed genetic groups based on population assignment from ADMIXTURE (Figure 2) to either the GL or BR population for each species. For rock bass, yellow perch, smallmouth bass, and white sucker, we assigned individuals to either the GL

group or the BR group by using a membership proportion cutoff of 0.5 from ADMIXTURE population assignment. For walleye, the PCA exhibited three distinct genetic groups (one BR population and two GL populations), and we assigned individuals to one of these three groups (BR,GL1, GL2) using a membership proportion cutoff of 0.33 from ADMIXTURE population assignment. We then recalculated summary statistics, F_{ST}, and N_e, and developed PCAs using these new genetic groups.

We used the program BAYESCAN (Foll & Gaggiotti, 2008) to identify outlier loci potentially displaying signals of directional selection in each species except walleye, which were likely stocked from multiple non-local sources (see results). BAYESCAN was run using the default parameters and a conservative false discovery rate of 0.01 to identify putative outliers. We then assessed the putative function of outliers by querying their sequences in the NCBI nucleotide and protein databases using BLAST. Only alignments with e-values < 10^{-10} were retained.

Comparison of empirical data to simulated migration scenarios

We simulated multiple demographic scenarios to investigate whether the levels of genetic differentiation that we observed between originally sampled above- and below-dam populations were consistent with recent isolation (i.e. isolation by the Union Street Dam) or longer-term isolation that existed before construction of the dam. We did not consider walleye in this analysis as they were likely stocked from multiple non-local sources (see Results). To approximate the construction of a dam, we simulated scenarios where a barrier was placed between two panmictic populations 30 generations ago. We chose a generation value of 30 as our species have a generation time of approximately 5 years (Becker, 1983; Scott & Crossman, 1985), and the Union Street Dam was built approximately 150 years ago. We then explored the influence of

different migration rates and N_es on genetic differentiation (F_{ST}). Scenarios were simulated 10 times each in the coalescent-based simulation in program fastsimcoal2 (Excoffier & Foll, 2011), and all simulations were initialized with two populations that initially exchanged a high number of migrants (m = 0.1 or a migration rate of 10%) before the barrier was constructed. Simulations approximated data from 15,000 unlinked loci with a maximum of two alleles and a mutation rate of 0.0001, and genetic statistics were assessed by sampling 50 individuals from each population. These parameters produced simulated datasets that were similar to the number of polymorphic loci (~5,000 loci polymorphic in each simulation) and levels of heterozygosity (~0.2) in our empirical datasets. We simulated data using two values of N_e (100, 1000) that approximated the resulting N_e for the species we analyzed (N_e = 100 was similar to rock bass and smallmouth bass and N_e = 1000 was similar to yellow perch and white sucker). For each N_e, we simulated a migration rate of zero and asymmetric migrations rates of zero upstream migration and downstream migration rates of 0.01, 0.05, and 0.1, or 1%, 5%, and 10%. We then used the R package diveRsity to estimate F_{ST} between simulated above and below dam populations and visualized the results with boxplots.

Relationships between genetic structure and ecological data

To assess any correlation between genetic structure and ecological attributes, we first compared total lengths of fish between populations within each species. We made comparisons between the originally sampled above- and below-dam populations and between the BR and GL genetic groups for each species. We evaluated test assumptions with the Shapiro-Wilk test for normality, visualization of Q-Q plots, and Levene's test for equality of variances. Length comparisons were performed using Student's 2-sample t-tests in instances where variances were equal, and by using Welch's unequal variances t-test when variances were unequal. For length

comparisons among walleye genetic groups (three populations rather than two), we performed a one-way ANOVA and then post-hoc Tukey's HSD test for multiple comparisons. Because we saw a mix of genetic groups within each sampling location for smallmouth bass, rock bass, and white sucker, we further split fish into groups for each combination of genetic group and sample location and compared lengths within these for each species using a one-way ANOVA and post-hoc Tukey's HSD after testing assumptions using the same methods listed above. For all tests, we used $\alpha=0.05$.

We also qualitatively assessed if the proportion of GL and BR fish within each sampling event varied across sample date because several collection events occurred for most species. We plotted the proportion of each genetic group within each sampling event for each species and visually assessed how proportions differed across dates. It is important to note that sampling was not performed evenly across time, sampling effort may not have been consistent between events, and some dates have very low sample sizes (as few as one fish in some instances), therefore we did not perform a statistical analysis for this assessment.

RESULTS

Quality control

A total of 428 individuals were RAD sequenced and 346 remained after filtering (Table S1). We successfully genotyped 49 to 86 individuals per species, with the smallest sample size per population in walleye, where we genotyped 22 individuals sampled above the dam and 27 individuals below (see Tables 2 and S1 for information on number of individuals sampled and genotyped). The average number of reads per individual was generally high and ranged from 7,768,406 in yellow perch to 14,511,218 in white sucker. The percent of individuals genotyped

per species ranged from 62% (walleye) to 89.6% (white sucker), with all species except walleye displaying a retention rate > 85%. The low genotyping rate in walleye was likely a function of low sample quality in the above dam samples. After filtering, missingness in individuals averaged 12.5% across all species, and ranged from 9.3% in walleye to 16.3% in white sucker (Table S1). The number of putative SNPs identified in each species ranged from 3,812 in yellow perch to 38,126 in white sucker with an average about 13,000 SNPs per species (Table 2).

Genetic differentiation and diversity

Visualization of ADMIXTURE plots (Figures 2 and S1) and PCAs (Figure 3) suggested unexpected genetic groupings not consistent with original population designations (above- and below-dam) for most species. The best supported number of ancestral populations (*K*) from ADMIXTURE was 2 for all species except white sucker (*K*=1). However, white sucker clearly split into two groups along the first principal component (PC1) in the PCA and the same split was apparent in the ADMIXTURE analysis at *K*=2. Therefore, we split white sucker into two genetic groups based on population assignment from ADMIXTURE as we did with rock bass, yellow perch, and smallmouth bass. For walleye, PCA and ADMIXTURE demonstrated three distinct groupings, with two GL groups separated from the BR group along PC1 and separated from each other along PC2. Although *K*=2 was the best fit according to ADMIXTURE cross-validation for walleye, we suspect this was because the upstream group was highly diverged from both downstream groups (see below), and retained the three genetic groups for plots and analyses.

After reforming populations using ADMIXTURE assignments, two general patterns became apparent across species: First, for rock bass, smallmouth bass, white sucker, and walleye, we generally found a single homogenous BR group above the dam and mixture of the BR group

with one or more other genetic groups of putative Great Lakes (GL) ancestry below the dam. Second, for yellow perch, we found distinct populations in Boardman Lake (BR group) and Traverse Bay (GL group), with no observed mixing below the dam.

Patterns in PCA plots varied among species (Figure 3). For rock bass, the BR group clustered relatively tightly, accounted for about 2/3 of individuals caught, and contained roughly equal proportions of fish sampled above and below the dam. The GL group separated from the BR group across PC1 and consisted only of fish caught below the dam. White sucker demonstrated the least amount of clear genetic separation of our study species, but two distinct groups split along PC1 and were consistent with our genetic groupings. The BR group contained individuals from both above and below the dam, while the GL group consisted mostly of fish caught below the dam but some were caught upstream as well; these fish clustered more closely with the BR group. Smallmouth bass appeared to split into BR and GL groups along PC1, with potentially one or two other intermediate genetic groups (or possibly hybrids) that we did not analyze separately due to low sample sizes. All GL individuals were caught below the dam except for one fish caught upstream. The BR group was larger and consisted of individuals both from below and above the dam. Yellow perch genetic groups were very similar to the originally sampled populations, with two clear groups split along PC1. Except for five individuals, genetic groupings based on ADMIXTURE aligned with the originally sampled populations; all fish caught upstream assigned to the BR group, and most fish caught in Grand Traverse Bay assigned to the GL group. As mentioned above, the walleye PCA showed three distinct groups which consistently aligned to their genetic assignment using admixture. Both GL groups consisted solely of individuals caught below the dam, while the BR group consisted of mostly upstream fish as well as five fish caught downstream that likely originated in Boardman Lake and passed

downstream through the dam. Investigation of the stocking history for this system suggest these three groups are likely the product of decades of stocking from multiple sources (GLFC, 2020; MIDNR, 2020). These records show Boardman Lake was stocked with walleyes from New York (now our BR group), and Grand Traverse Bay has been stocked repeatedly from two main sources, Muskegon and Little Bay de Noc (now our GL1 and GL2 groups), which are both Lake Michigan systems but are located on opposite sides of the lake from each other. No genetically intermediate individuals were observed in our walleye populations, even between the two GL populations, suggesting that these groups do not interbreed or that successful reproduction is not occurring below the dam where populations are mixed.

Genetic differentiation (F_{ST}) for all species was highly significant in both original population groupings (i.e. above vs below the dam) and population groupings based on ADMIXTURE analysis. The F_{ST} of original populations ranged from 0.0046 (white sucker) to 0.0671 (walleye) (Table 2) but increased dramatically when we reformed genetic groups. Rock bass had a large increase, from 0.0261 to 0.1021, as did smallmouth bass which changed from 0.063 to 0.1326. White sucker F_{ST} was the lowest of all species in both scenarios and went from 0.0046 to 0.0092. Yellow perch had the smallest change, from 0.0346 to 0.0371, likely because population assignment did not change as drastically as our other study species. Walleye F_{ST} changed from 0.0671 (pairwise) to 0.1053 (overall). Because we changed walleye from the original two populations to three genetic groups, we assessed pairwise F_{ST} between the three groups in addition to overall F_{ST}. The two GL groups had a pairwise F_{ST} of 0.057, and the BR group had a pairwise F_{ST} of 0.112 and 0.118 between GL1 and GL2, respectively, providing further evidence to support our hypothesis that the BR group is derived from a highly divergent out-of-basin source (New York).

Observed heterozygosity (H_o) differed by 0.01 or less between original above- and below-dam populations for all species (Table 2). When we reassigned populations to the BR and GL genetic groups, heterozygosity again differed by 0.01 or less for all species except smallmouth bass, where H_o was 0.276 for the GL group and 0.306 for the BR group. Allelic richness demonstrated similar patterns, with little variation among groups outside of regrouped smallmouth bass populations, where allelic richness differed by ~ 0.05 between the GL and BR groups. Estimates of F_{IS} were near zero (between -0.04 and 0.02) for all populations grouped based on ADMIXTURE except for white sucker, where F_{IS} was ~0.12 in all populations for all groupings.

We conducted a number of analyses to attempt to understand why F_{IS} was elevated in white sucker and hypothesize that this trend was the result of a genome duplication in Catostomidae (Uyeno & Smith, 1972). Specifically, we first tested different -M (mismatches allowed) values in STACKs; values tested were 1, 3 (original value), and 7. The pattern of high F_{IS} was present at all -M values. We then conducted analyses outside of STACKs by aligning quality filtered reads for sequenced individuals to a small number of loci with high F_{IS} values and compared read counts and genotypes derived from STACKs with these alignments. We found that many loci had a large number of reads for a sequence that was a close or exact match to the target allele and a sequence that was similar but substantially different (potentially a paralog). Many of these loci were called as homozygotes in STACKs, leading to high frequencies of alternate homozygotes, few heterozygotes, and high F_{IS} values. Unfortunately, few genomic resources exist for Catostomidae, making confirming our hypotheses difficult, but we hope that sequenced genomes for this family will clarify the pattern in the future. We do, however, believe that analyses based on our full dataset are robust, as we analyzed population

structure with three different datasets (loci with $F_{IS} < 0.2$, $F_{IS} < 0$, and $F_{IS} > 0$) and found the same patterns (F_{ST} values within 0.0006 of values in the overall datasets and extremely similar patterns of population divergence in PCAs). We therefore decided to retain our full dataset for all analyses with the caveat that true F_{IS} values may be lower than our estimates.

Estimates of N_e ranged from 14 to over 12,000 and generally increased when populations were grouped according to ADMIXTURE rather than sampling location (Table 2). Rock bass and smallmouth bass tended to have the lowest N_e estimates, between 75 and 192 when populations were grouped by ADMIXTURE. White sucker and yellow perch had much larger N_e estimates near or above 1,000 for all population groupings, with estimates as high as 8,845 for yellow perch samples taken above the dam. Estimates of N_e for walleye varied substantially between the three genetic groups defined by ADMIXTURE, with estimates of 221 for the BR group, 1492 for the GL1 group, and 26 for the GL2 group. It is important to note that two related pairs of individuals were found in the GL2 group (i.e. four individuals out of 10 were related, see below) and this may have potentially reduced the N_e estimate for this group. Estimates of N_e were generally similar between BR and GL groups, but slightly higher estimates were observed in the BR group for smallmouth bass, white sucker, yellow perch, and a slightly lower estimate was observed in the BR group for rock bass. Alignments to the yellow perch genome were successful for 1,970 tags (44%) in walleye and 3,746 tags (98%) in yellow perch; these alignments were used to remove bias due to physical linkage when calculating N_e for these species.

Related individuals (parent offspring or full siblings) were relatively rare in our dataset, with yellow perch and white sucker containing zero related pairs, walleye containing two, and smallmouth bass and rock bass each containing four (Table S2). Related pairs were always

captured in the same sampling area (i.e. above or below the dam) and belonged to the same genetic group. In walleye, both related pairs were sampled downstream and belonged to group GL2. In smallmouth bass three of four pairs were sampled above the dam and belonged to the BR group, and the other pair was sampled below the dam and belonged to the GL group. Three of four rock bass pairs were sampled below the dam, one was sampled above the dam, but all pairs assigned to the BR group. Most related pairs appeared to be siblings as they were generally similar in length, but three of four pairs in smallmouth bass and one of four pairs in rock bass differed substantially in length and may have represented parent-offspring pairs. The number of related individuals in each species appeared to be somewhat related to N_es, as species with larger N_es (white sucker, yellow perch) did not have any related individuals. We retained related pairs for all analyses as we have no reason to believe that rates of relatedness that we observed are non-representative of each population (Waples & Anderson, 2017).

Outlier tests identified a relatively small number of highly differentiated loci: zero loci in yellow perch, two in smallmouth bass and rock bass, and 11 in white sucker (Table 2, Figure 4). In general, distributions of F_{ST} were relatively continuous and did not reveal large breaks with highly differentiated loci. Additionally, the fact that only two populations were included in each analysis likely led to low power for detecting outliers. We were able to successfully align four out of 15 outliers to protein sequences, one locus for smallmouth bass, one locus for rock bass, and two loci for white sucker (Table S3). Our most notable alignment was the locus in rock bass, which aligned to an immunoglobulin-like protein that may be involved in immune system function. The other loci aligned to a transposable element, and genes coding for an integrase and elongation factor (Table S3).

Comparison of empirical data to simulated migration scenarios

Simulations of zero and asymmetric migration (m) revealed that after 30 generations, two populations with N_es of 100 should display F_{ST} values averaging 0.145, 0.109, 0.046, and 0.025 under migration rates of 0, 0.01, 0.05, and 0.1, respectively (Figure 5a). The two species with N_e close to 100, rock bass and smallmouth bass, displayed F_{ST} values of 0.13 and 0.10 respectively, which most closely match simulations with either no or very low (m=0.01) migration. This indicates that if the population differentiation that we observed was caused by the Union St. Dam, migration between above and below dam populations would need to be extremely small. However, we observed a large number of individuals of both GL and BR origin below the dam. We therefore suggest that the most probable explanation for the patterns of genetic structure that we observed in rock bass and smallmouth bass is the existence of separate BR and GL populations, with GL populations mixing with BR populations in the lower river but rarely interbreeding.

Simulations of two populations with N_es of 1,000 unsurprisingly produced smaller F_{ST} values, with averaging 0.016, 0.012, 0.005, 0.003 under migration rates of 0, 0.01, 0.05, and 0.1, respectively (Figure 5b). The two species with N_es near 1,000, white sucker and yellow perch, displayed F_{ST} values of 0.009 and 0.037 respectively. It is plausible that genetic differentiation similar to the levels observed in white sucker could occur in the timeframe since the dam was built if migration was low (observed F_{ST} for white sucker similar to the value observed for m=0.01). However, the fact that BR origin fish are highly mixed in our samples from the lower river suggest that downstream migration could be much higher than 1%, yet significant differentiation persists, suggesting that BR and GL fish do not interbreed frequently. The F_{ST} value for yellow perch (0.0371) was over two times higher than the F_{ST} estimate for the m=0

scenario; yellow perch therefore provide the strongest evidence for differentiation before dam construction among our study species.

Relationships between genetic structure and ecological data

Lengths of fish were significantly different between BR and GL groups defined by ADMIXTURE for all species (Figure 6). Walleye in the GL2 group were significantly longer than the other two groups but there was no significant difference between the lengths of GL1 and BR groups. For yellow perch, the GL group was longer than the BR group, but for the other three species (smallmouth bass, rock bass, and white sucker), fish from the GL group were shorter than those from the BR group. The most striking differences in lengths were in smallmouth bass, where fish from the GL group were 151 mm shorter on average compared to the BR group (Figure 6, Table 2). Differences were smaller for other species ranging from ~30 mm in rock bass to ~55 mm in white sucker. Two of the three walleye groups did have large differences in lengths, with GL2 averaging 662 mm and BR averaging 430 mm (a difference of ~230 mm), but we suspect this was because fish in GL2 were primarily large spawning fish that were sampled in the river below the dam.

For length comparisons among combinations of genetic group and sample location, the GL-downstream white suckers were significantly shorter than both the BR-downstream and the BR-upstream groups (Figure S2). The same pattern held true for rock bass. As above, the most striking differences in lengths were observed in smallmouth bass. Both the BR and GL fish caught below the dam were significantly shorter on average (222 mm and 182 mm, respectively) compared to the BR fish caught upstream (405 mm). Within both genetic groups caught below the dam, however, there were a small number of larger fish that were of spawning-size (350 -

455 mm; Figure S2). We did not make comparisons with the GL-upstream group of smallmouth bass as it contained a sample size of 1.

For most species, no clear patterns emerged when we visually assessed the proportion of each genetic group by sampling date other than relatively higher numbers of fish caught below the dam in general during months where spawning was likely occurring (Figure S3). For example, 36 white sucker were caught below the dam in April 2019, and only 5 were caught in June 2019. In general, however, the number of fish sampled on each date was either low (walleye), clustered into one major sampling effort per population (yellow perch), or proportions of fish in each genetic group were similar (rock bass and white sucker). For smallmouth bass, however, it appeared that GL fish made up a greater proportion of the fish sampled below the dam during May 2017 compared to July 2017 or September 2018, which each had a comparatively higher proportion of BR fish, but lower numbers of fish overall (Figure S3).

DISCUSSION

While dams are ubiquitous worldwide, few studies have assessed the genetic structure of multiple fish species across a small spatial scale (<5 km) in a dammed river. Our results demonstrate significant genetic differentiation on the scale of a few km within all five of our study species (rock bass, white sucker, smallmouth bass, yellow perch, and walleye) in the fragmented lower section of the Boardman River. However, the genetic structure we observed could not be explained solely by the presence of the Union Street Dam. For most of our study species (except for walleye, which have been extensively stocked from different basins), our PCA, admixture, and migration simulations suggest that historical subpopulations, which likely diverged long before the construction of the dam, have existed in the lower Boardman River, overlapping occasionally in time and space, but maintaining reproductive isolation. We suggest

that individuals from a Great Lakes (GL) genetic group, which has adapted to a larger lake system, specifically Grand Traverse Bay and Lake Michigan, enter the lower Boardman River in certain life stages and coexist with individuals from the Boardman River (BR) genetic group. Before the Union Street Dam was constructed, both BR and GL fish within our study species were able to move freely throughout the Boardman River watershed and Grand Traverse Bay, and each genetic group likely used these different habitats during specific times of the year or during different stages of growth and development. Without our genetic assessment, the overlap of these distinct genetic groups within our study species in the lower Boardman River would have likely continued undetected. These results also highlight the importance of tributary and river mouth habitats for preserving fish populations in the Great Lakes and emphasize that a conservation approach centered around maintaining high connectivity and habitat quality of tributaries is vital for ecosystem health.

Longstanding genetic differentiation prior to dam construction

While geographic isolation is often the largest factor influencing the landscape of genetic differentiation, research has highlighted that isolation by distance (IBD) may not always be the main variable driving contemporary population genetic structure. For example, adjacent populations may experience isolation by adaptation (IBA), in which gene flow is reduced as a result of local adaptation along differing ecological or environmental gradients (Nosil et al., 2009; Orsini et al., 2013; Rasanen & Hendry, 2008). IBA can occur across small spatial scales in the absence of geographic barriers to dispersal, revealing cryptic and often unexpected genetic structure that does not necessarily follow an IBD pattern. For example, Nosil et al. (2008) discovered that ecotypes of walking stick insects (*Timema cristinae*) adapted to different host species exhibit IBA at both neutral loci and loci under putative selection. IBA has also been

documented in aquatic systems, including a study by Bond et al. (2014), who detected strong differentiation between freshwater and estuarine Dolly Varden (*Salvelinus malma*) populations within a small watershed lacking physical barriers, and hypothesized that IBA due to varying selective pressures across different environments was the main driver of observed structure.

We speculate that genetic differentiation between BR and GL populations in our study was most likely driven by IBA after deglaciation in the region, long before construction of the Union Street Dam. Great Lake and tributary habitats differ in a number of attributes including temperature, nutrient levels, and composition of prey species (Bhagat & Ruetz, 2011; Brazner & Beals, 1997; Höök et al., 2007; Janetski & Ruetz, 2015), likely leading to groups of fish that display high adaptive divergence. Other recent studies throughout the Great Lakes have also revealed fine-scale genetic structure between fish populations inhabiting main lakes and tributaries, suggesting that historical divergence through IBA may be more common within the Great Lakes and their tributaries than previously thought. For example, Euclide et al. (2020) detected strong genetic structure in Lake Michigan smallmouth bass across small spatial scales (10-30 km), which often correlated to differences in habitat type rather than geographic distance, and found that gene flow between lake and river sites was low, even though individuals from the two habitat types likely mixed outside of the spawning season. Additionally, Chorak et al. (2019) observed genetic structure in yellow perch populations between Lake Michigan and connected DRM habitats, even though the different populations overlapped during parts of the year. These findings support our hypothesis that distinct BR and GL populations of our study species may have adapted to different habitats types (Great Lake vs tributary) and historically maintained low levels of gene flow without the presence of barriers.

The high differentiation observed between BR and GL populations and the fact that our outlier tests resulted in few or no loci under selection suggests that BR and GL populations have substantially diverged across the genome. Research suggests that populations that diverge without barriers to gene flow such as when IBA occurs should demonstrate highly heterogeneous patterns of differentiation across the genome (i.e. genomic islands of divergence; Nosil & Feder, 2012; Via, 2012). However, as divergence time increases, differentiation at neutral loci should increase as well, potentially obscuring signatures of selection (Larson et al., 2019; Nosil et al., 2008; Via, 2012). We observed no conspicuous peaks of potentially adaptive loci in any of our species, providing further evidence that GL and BR populations have been diverged for a substantial amount of time. We speculate that islands of divergence may have existed in our species when IBA was first occurring, but that genome-wide differentiation has become large enough to obscure them.

After initial IBA occurs, other mechanisms, such as natal homing and site fidelity, can help to increase and maintain differentiation (Lin et al., 2008), and we suspect this is occurring in our study system as well. Both rock bass and smallmouth bass tend to occupy relatively small home ranges and exhibit spawning site fidelity (Gerber & Haynes, 1988; MacLean & Teleki, 1977; Ridgway et al., 1991), which could increase or maintain differentiation that was initially created by IBA. White suckers exhibit distinct spawning migrations into tributaries in the spring to spawn, and also display natal homing (Doherty et al., 2010; Geen et al., 1966; Werner, 1979). Before the construction of the Union Street Dam, the resident BR population of white sucker likely spawned in tributaries upstream in the watershed, while the GL fish may have spawned lower in the drainage or in Kid's Creek, a small tributary that enters the river a few hundred meters below the dam, thus reducing gene flow with the BR group. The differentiation we

observed between the white sucker genetic groups support this hypothesis. Yellow perch are primarily a lentic species and generally do not exhibit spawning migrations intro tributaries like our other species, but do exhibit broad natal homing within their resident system, which can lead to genetic differentiation between spawning groups (Leung & Magnan, 2011; Parker et al., 2009; Sepulveda-Villet et al., 2011). Our results suggest that the BR and GL groups of yellow perch consistently reproduce within their respective lake systems and thus remain genetically divergent.

Although fragmentation by artificial barriers can impact the genetic structure and diversity of fish populations, especially in migratory species with strong natal homing like salmonids (Horreo et al., 2011; Samarasin et al., 2017; Wofford et al., 2005; Yamamoto et al., 2004), our data do not suggest that the Union Street Dam is the primary driver of genetic structure in our system. The construction of the dam happened relatively recently in evolutionary terms (~150 years ago). To observe any genetic impact over the lifetime of this dam, affected fish populations would need to be relatively small and there would have to be essentially zero downstream gene flow through the dam (as seen in our simulation results and also suggested by Hoffman et al., 2017; Keyghobadi, 2007; Selkoe et al., 2015). For most of our study species, our simulations demonstrated that if each species consisted of one genetically homogenous population before dam construction, a migration rate of 0% or 1% would have been necessary to produce the F_{ST}s that we observed. However, based on the high mixing of the two populations below the dam for most species, and the likelihood that significant downstream movement of juveniles through the dam occurs (as seen in the BR walleye caught below the dam), a nonexistent or extremely low level of downstream gene flow seems highly unlikely in this system, and therefore the dam could not have caused the differentiation we observed. Yellow

perch was the only species that did not demonstrate mixing of the two groups below the dam, but the F_{ST} for that species was twice as high as the resulting simulated F_{ST} under a 0% migration rate, meaning the two populations could not have been homogenous at dam construction.

Ecological differences suggest ontogenetic shifts

Our results suggest that the different genetic groups we detected experience ontogenetic habitat shifts with movement between the BR and GL systems. Specifically, we hypothesize that juvenile fish from the GL group enter the lower Boardman River from Grand Traverse Bay for a period of time, likely to seek food and/or refuge. Additionally, some BR fish may also leave the river and enter the bay during parts of the year as well, as many species are not captured consistently in the river below the dam throughout the year (R. Swanson, GLFC, personal communication). For most of our study species, GL fish in the lower Boardman River were significantly smaller than the BR fish that they were mixing with. However, it is important to note that our length data must be interpreted with caution due to gear selectivity, variable capture dates, and a lack of age data; nevertheless, the trends we observed are unlikely to be statistical artifacts as they were highly consistent across species with different size ranges. Data on length and sampling date are available for many genetic studies, but these data are rarely incorporated into conservation genomics research. We thus demonstrate the utility of incorporating ecological data to gain a clearer picture of fish life history and movement patterns, and we suggest that conservation genomic studies should explore incorporating this type of data more frequently.

Most fish species demonstrate ontogenetic niche shifts, switching to different food sources or habitats during different life history stages, and movement to nearshore, wetland, or tributary habitats represent a common ontogenetic shift for growing fishes (Persson & Crowder, 1998; Werner & Gilliam, 1984). In Lake Michigan, river mouths and tributaries are unique

ecosystems that harbor diverse and variable fish assemblages (Janetski & Ruetz, 2015; Larson et al., 2013). These systems are generally characterized by relatively warmer temperatures, higher productivity and turbidity, and more macrophyte cover compared to the larger lake to which they are connected (Höök et al., 2007; Larson et al., 2013). Consequently, these habitats are important nursery and refuge areas for juvenile fish (Altenritter et al., 2013; Brazner & Beals, 1997; Madenjian et al., 2018). In our study, evidence for an ontogenetic habitat shift of young GL fish into the lower Boardman River was strongest for smallmouth bass. While this species is often considered relatively sedentary, Humston et al. (2017) found that young smallmouth bass exhibited a high degree movement between river systems and tributaries, with some age-0 fish traveling at least several km from their natal site, and suggested that differences in lake and river habitats could be driving this dispersal. Similar variations in life history strategies have recently been uncovered in our other study species as well. Chorak et al. (2019) and Senegal et al. (2020) both found that in eastern Lake Michigan, yellow perch exhibited multiple life history variations, in which some yellow perch were Lake Michigan residents, some were DRM lake residents, and some were Lake Michigan fish that temporarily moved into DRM lakes during the fall. It is therefore likely that prior to dam construction, GL juvenile smallmouth bass, rock bass, yellow perch, and white sucker (to some degree) not only entered the lower Boardman River, as we observed in our study, but also traveled upstream of the current dam site and utilized habitat in Boardman Lake, which is a natural DRM lake, and perhaps further upstream as well.

Conservation/management implications and conclusions

We have demonstrated that the genetic structure of our five study species in the Boardman River is not a result of fragmentation by the Union Street Dam, but a consequence of historical divergence of two distinct subpopulations, one of which developed a Great Lakes life

history and the other a tributary life history, that sometimes overlap in distribution but remain reproductively isolated. Without our genetic assessment, the presence of these cryptic ecotypes that exist across small spatial scales in the lower Boardman River would have continued undetected. Our findings highlight the fact that fish from genetically distinct groups may sometimes overlap spatiotemporally but have distinctive life histories with unique population dynamics and habitat requirements during different life stages. These multiple life histories are likely important components of a robust portfolio of within-species diversity that promotes population stability and resilience in the face of environmental stochasticity (Schindler et al., 2010).

Even if river fragmentation does not result in population declines or drastic genetic impacts as some studies have observed (e.g. Horreo et al., 2011; Raeymaekers et al., 2008; Yamamoto et al., 2004), the restoration of connectivity between lake and tributary habitat is still essential for conservation and restoration of fish populations. Connectivity between Lake Michigan, Boardman Lake, and the entire Boardman watershed was likely historically important for various components of the life histories of all our study species. Species that migrate from the Great Lakes into tributaries to spawn like white sucker and walleye are more obviously impacted by barriers, but our research suggests that species like rock bass, smallmouth bass, and yellow perch, that may not exhibit similarly distinct spawning runs over long distances, may also be negatively impacted by fragmentation. Even if their utilization of river habitat is less obvious, populations of these species in Lake Michigan likely rely upon tributary and DRM habitats, especially as juveniles, for feeding and refuge. The role of tributary and DRM habitat in Great Lakes fishery production has been underappreciated, but studies like ours are illuminating the importance of protecting these unique ecosystems feeding into the Great Lakes, which play a

vital role in fish recruitment, growth, and reproduction. Based on our results, we suggest that fisheries managers in the Great Lakes and beyond adopt a holistic viewpoint of fish populations that takes into account the existence of unique and partially sympatric genetic groups as well as the importance of habitat connectivity across multiple life stages.

In conclusion, our study combined genetic and ecological data to illuminate cryptic population diversity that could have major implications for how fish populations in the Great Lakes are managed. Additionally, our workflow which included ecological data, genetic data, and simulations can be applied to other systems to investigate the relative importance of dams and other factors in shaping population structure. In conjunction with traditional survey methods, genetics has the power to elucidate cryptic and underappreciated diversity. We suggest that resource managers seek to incorporate genetic analysis into their toolbox more frequently to better understand and conserve important habitats and populations.

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Table 1: Summary of general life history traits of our study species in the upper Great Lakes region. Life history traits were derived from Becker (1983), Scott and Crossman (1985), and input from regional biologists (Daniel Zielinski GLFC, Dan Isermann USGS, personal communication).

Species	Reproductive Strategy	Migration Distance	Spawning Philopatry	Life Span (yrs)	Age at Maturity (yrs)	Fecundity
Rock Bass	Nest	Short	Med	6-10	2-3	Low - Intermediate
White Sucker	Broadcast	Med-Long	Med-High	~20	2-8	Intermediate - High
Smallmouth Bass	Nest	Short-Med	Med-High	~18	2-4	Intermediate
Yellow Perch	Broadcast	Med	Low-Med	8-12	2-4	Intermediate
Walleye	Broadcast	Med-Long	Med-High	5-20	2-6	High

Table 2: Summary of RADSeq data, population information, and genetic diversity metrics calculated for our five study species in the lower Boardman River. Summary statistics were calculated both for original sample populations (below dam and above dam), and for putative genetic groups assigned from ADMIXTURE (Great Lakes group and Boardman River group). N SNPs is the total number of putative SNPs detected within each species after filtering. N outlier loci are the number of highly differentiated loci within each species, related pairs are the number of pairs within a species with a Wang relatedness estimation of 0.4 or higher (siblings or parent/offspring). Sample size is the number of individuals successfully genotyped in a given group. F_{ST} is the measure of genetic differentiation, AR is allelic richness, H_O is observed heterozygosity, H_E is expected heterozygosity, F_{IS} is the inbreeding coefficient, N_e is effective population size. Avg TL is the average total length of individuals within a group.

Species	N SNPs	N outlier loci	Related pairs	Group	sample size	F_{ST}	AR	H_o	H_e	F_{IS}	N_e	N_e CI Low	N_e CI High	Avg TL (mm)	SD
				Original population											
Rock Bass	8655	2	4	**Below dam**	41	0.0261	1.983	0.317	0.341	0.063	24.9	24.9	24.9	154	40.6
				Above dam	27		1.978	0.315	0.32	0.01	78.2	77.5	78.8	164	31.8
				Genetic group from Admixture											
				Great Lake	20	0.1021	1.935	0.316	0.32	0.01	196.9	191.9	202.2	135	36.6
				B. River	48		1.964	0.316	0.322	0.016	104.9	104.3	105.5	167	33.8
				Original population											
White Sucker	38126	11	0	**Below dam**	44	0.0046	1.987	0.258	0.298	0.12	1246.7	1230.2	1263.7	443	54.3
				Above dam	42		1.989	0.257	0.298	0.122	693.1	687.6	698.7	477	68.2
				Genetic group from Admixture											
				Great Lake	32	0.0092	1.965	0.255	0.255	0.118	831.3	819.7	843.2	426	54.3
				B. River	54		1.982	0.26	0.299	0.121	1117.9	1107.3	1128.8	480	60.2
				Original population											
Smallmouth Bass	11044	2	4	**Below dam**	33	0.063	1.981	0.291	0.308	0.049	14.1	14.1	14.2	199	95.8
				Above dam	27		1.967	0.301	0.307	0.019	66.4	66.0	66.8	404	87.4
				Genetic group from Admixture											
				Great Lake	20	0.1326	1.887	0.276	0.269	-0.021	75.3	74.5	76.1	192	112.5
				B. River	40		1.962	0.306	0.312	0.02	103.0	102.5	103.6	341	121.9
				Original population											
Yellow Perch	3812	0	0	**Below dam**	46	0.0346	1.935	0.265	0.266	0.004	801.3	738.4	875.8	253	28.4
				Above dam	37		1.964	0.268	0.27	0.007	8444.9	4066.2	∞	209	60.3
				Genetic group from Admixture											
				Great Lake	41	0.0371	1.925	0.264	0.263	-0.003	1977.1	1599.2	2587.1	254	28.1
				B. River	42		1.972	0.269	0.272	0.011	8250.8	4269	118434	213	58.5
				Original population											
Walleye	4470	NA	2	**Below dam**	27	0.0671	1.964	0.26	0.278	0.054	41.1	40.4	41.8	574	141.8
				Above dam	22		1.887	0.26	0.262	0.001	142.8	133.1	154	397	70.8
				Genetic group from Admixture											
				Great Lake 1	12	0.1053*	1.81	0.262	0.259	-0.02	1492.2	898.5	4370	495	171.7
				Great Lake 2	10		1.787	0.259	0.249	-0.04	26.0	25.6	26.4	662	68.6
				B. River	27		1.815	0.259	0.263	0.009	221.0	213.4	229.1	430	99.1

*Pairwise *F*ST values among walleye groups: GL1 - GL2 = 0.057, BR - GL1 = 0.112, BR - GL2 = 0.118.

Figure 1: Sampling area in the lower Boardman River in Traverse City, MI, USA which drains into northeast Lake Michigan. All "Upstream" sampled fish were collected in Boardman Lake, above the Union Street Dam. "Downstream" sampled fish were all captured in the Boardman River below the Union Street Dam, except for yellow perch which were collected from the West Arm of Grand Traverse Bay < 1 km from the mouth of the river.

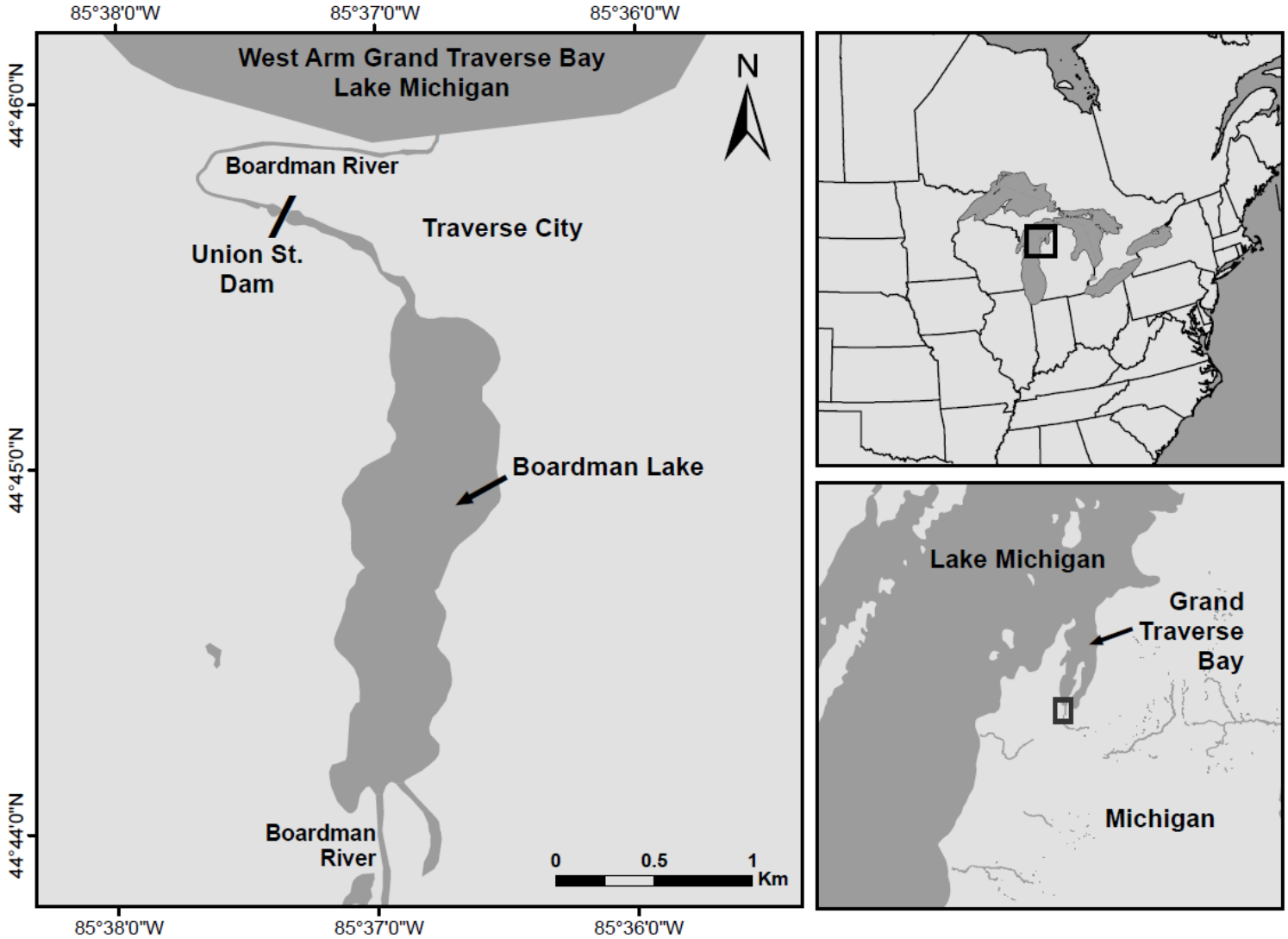


Figure 2: Genetic ancestry of all five study species captured either upstream or downstream of the Union Street Dam estimated with ADMIXTURE. Each vertical bar represents an individual, and color corresponds to ancestry proportions. All species (A) - (D) were best represented by two lineages (*K*=2), except for walleye (E) which was best represented by three lineages (*K*=3). Blue portions of ancestry correspond to the putative "Boardman River" (BR) genetic group, and red portions correspond to the putative "Great Lakes" (GL) genetic group. Walleye additionally split into a second "Great Lakes" group (GL2) which corresponds to the purple portion of ancestry.

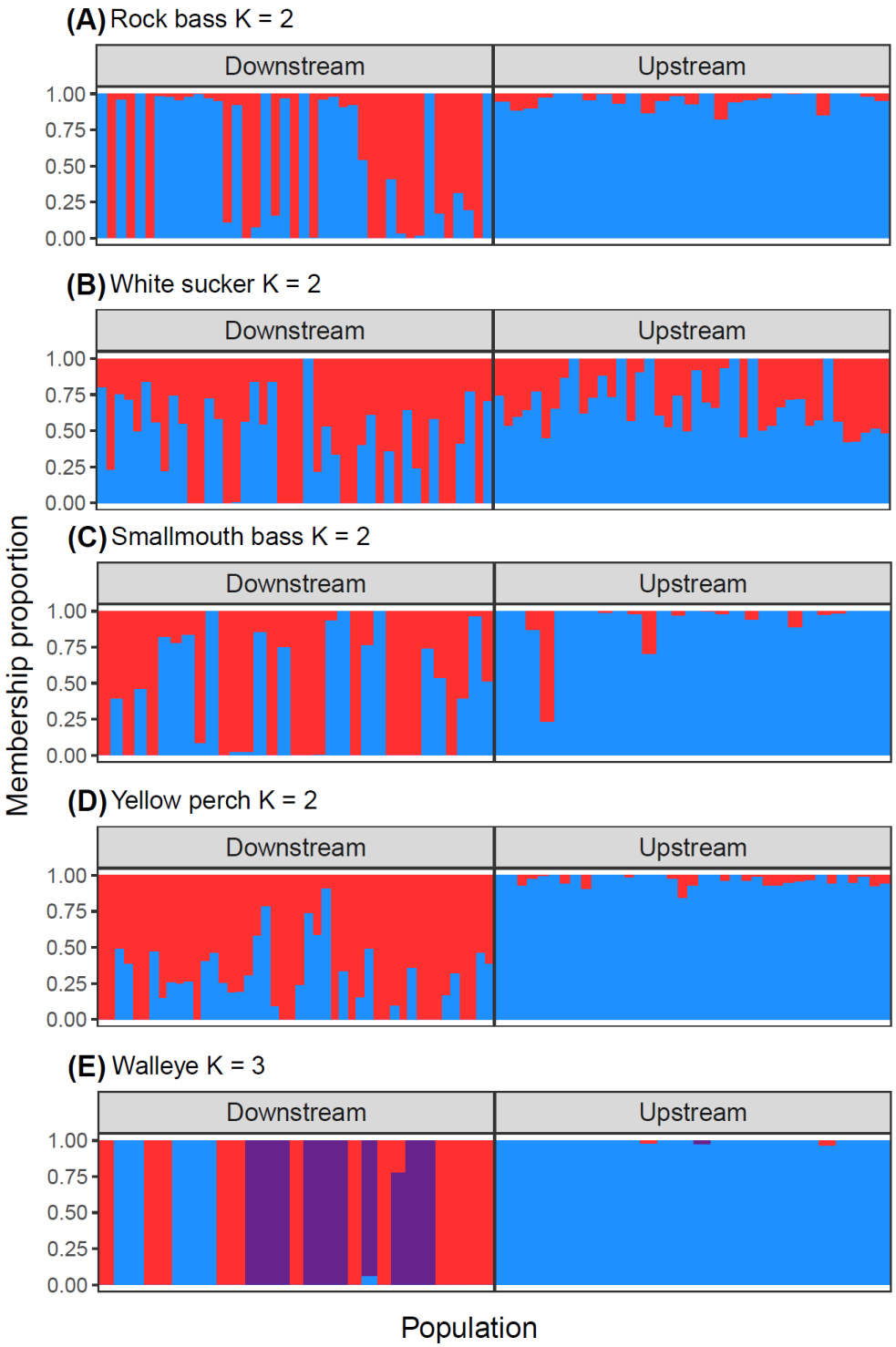


Figure 3: Principal component analysis (PCA) plots for all five study species (pictured in each plot). The percentage of variance explained by PC1 and PC2 is labeled on the x- and y- axes, respectively. Colors correspond to putative genetic group determined by ADMIXTURE assignment, with blue corresponding to the Boardman River (BR) group and red corresponding to the Great Lakes (GL) group. Walleye split into a third genetic group (GL2 in purple), which appears to be associated with the Great Lakes but is distinct from the first GL group. Circles correspond to individuals captured downstream of Union Street Dam, and triangles correspond to individuals caught above the dam. Fish illustrations were created by Joseph Tomelleri and used with permission.

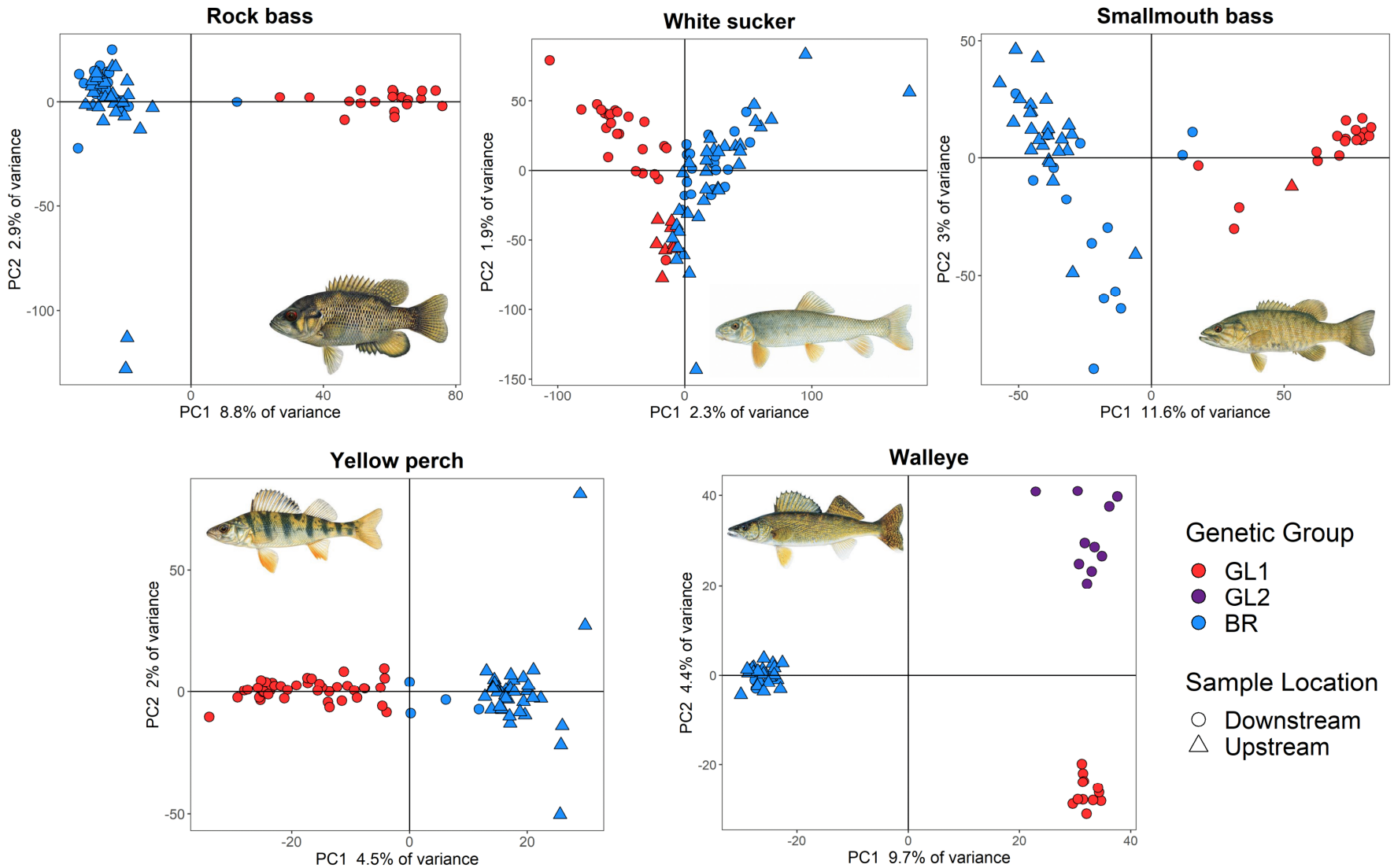


Figure 4: Visualization of putative outlier loci identified with BAYESCAN for a) white sucker, b) rock bass, c) smallmouth bass, and d) yellow perch sampled in the lower Boardman River. Loci in red are putative outliers (FDR correct probability < 0.01), and a vertical black line was drawn at this significance value. Walleye was not included for this analysis because populations have been heavily influenced by stocking.

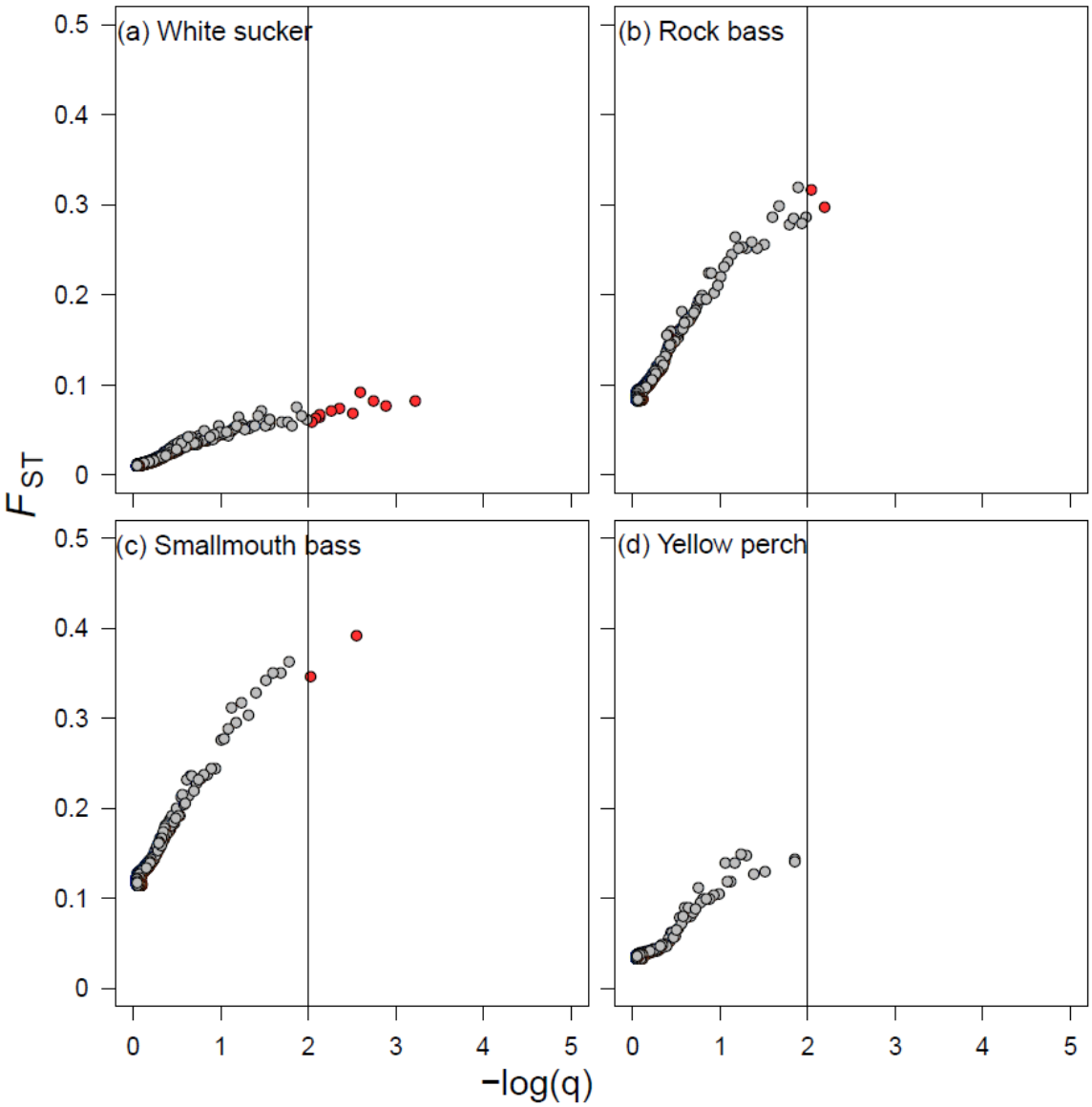


Figure 5: Results from simulated migration scenarios estimating genetic differentiation (F_{ST}) across a barrier over the approximate lifespan of the Union Street Dam (30 generations with an estimated generation time of 5 years). We tested populations with N_es of 100 and 1000, and migration rates of 0, 0.01, 0.05, and 0.1, simulating each scenario 10 times and plotting resulting F_{ST} outputs. Migration was fully asymmetric (upstream to downstream) in all scenarios, with no upstream migration.

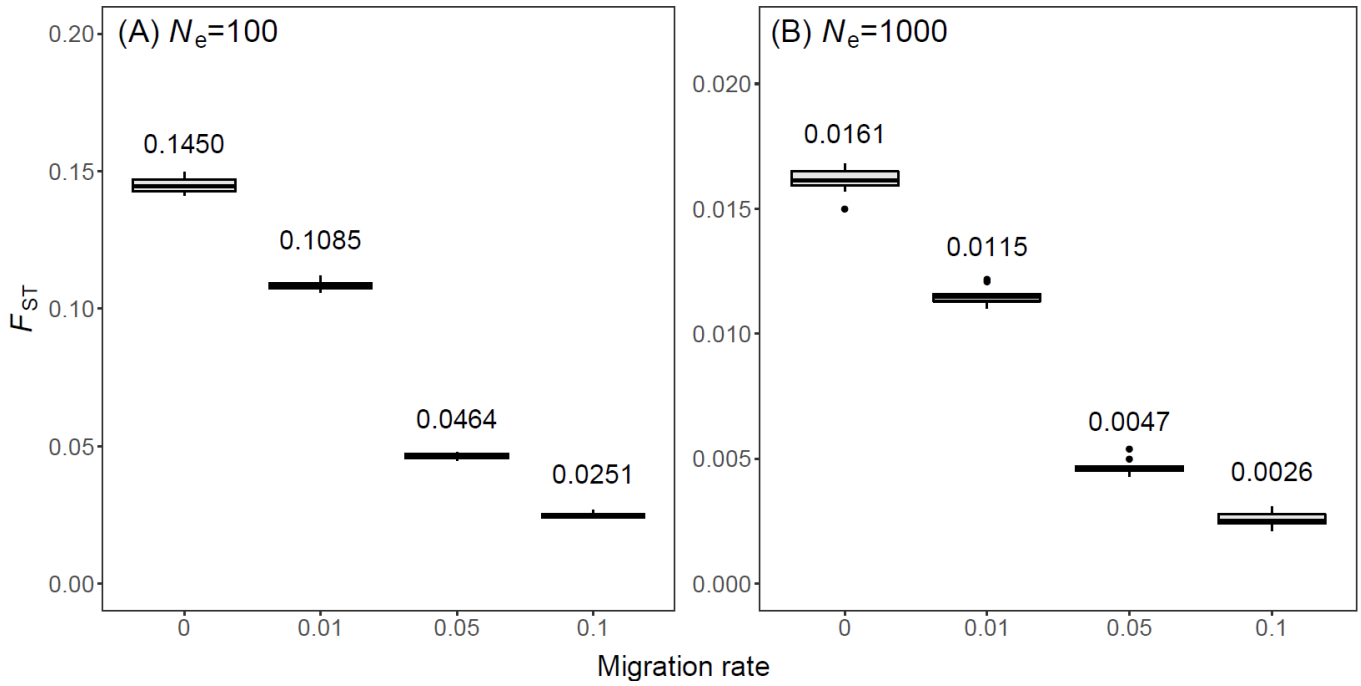
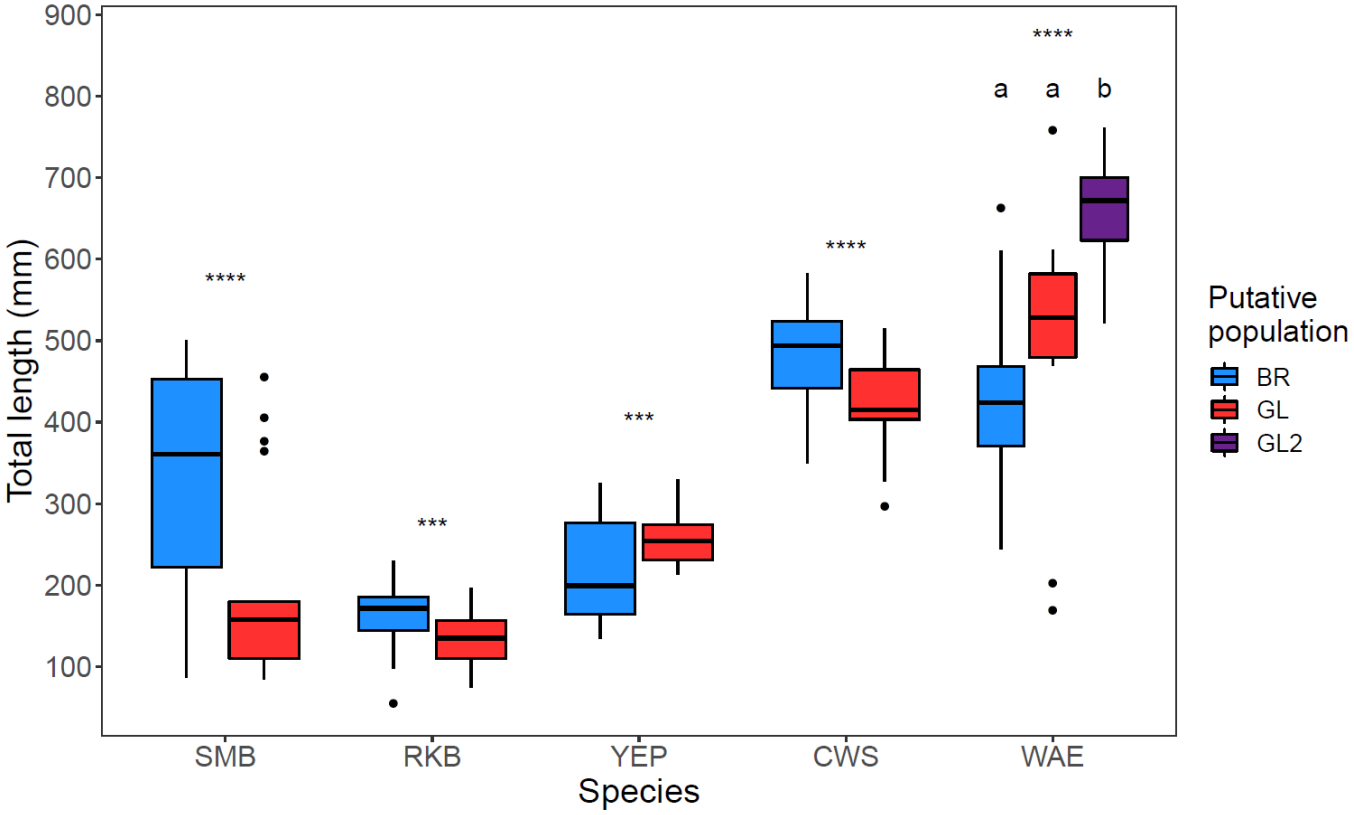


Figure 6: Differences in total length in mm between putative genetic groups within each study species in the lower Boardman River. All species were assessed using t-tests except for walleye, where we used ANOVA and Tukey's HSD as this species contained three genetic groups. All comparisons were significantly different at $\alpha = 0.05$.
* = $P \leq 0.05$, ** = $P \leq 0.01$, *** = $P \leq 0.001$, **** = $P \leq 0.0001$



SUPPLEMENTARY MATERIALS

Supplementary tables and figures can be found in the Chapter 1 folder of the following shared Google Drive which can be accessed using the link below:

https://drive.google.com/drive/folders/1eRaQilvMqp6elFMC-EzZ7fL43_s-ghoU?usp=sharing

Table S1: Total retained reads, average number of reads, and standard deviation (SD) of reads after filtering for each study species captured in the lower Boardman River sequenced with RADSeq. n attempted refers to the total number of original individuals sequenced within each species and within each population (Above or Below the Union Street Dam). n genotyped is the number of individuals after passing through QC filtering steps, and % genotyped is the % of individuals within each species/population that passed QC filtering. We also calculated mean % missingness of data per individual after filtering, along with standard deviation (SD), and the range of % missingness within each species.

Table S2: Estimated related pairs within each species across populations in the lower Boardman River. A pair of individuals were considered related if they had a Wang relatedness estimation of 0.4 or higher. The two entries in the related pair column correspond to the original sample codes for the individual fish. Genetic group corresponds to BR for Boardman River or GL for Great Lakes. Sample location refers to the originally sample population (Above or Below dam). Date of capture and total length in mm is also included for each individual.

Table S3: Results from querying putative outlier loci in the NCBI nucleotide and protein database using BLAST. Walleye was not included due to previous stocking from multiple sources, and yellow perch had no outlier loci.

Figure S1: Genetic lineages for all five study species captured either upstream or downstream of the Union Street Dam estimated with ADMIXTURE. We tested two to five lineages ($K = 2$, $K = 3$, $K = 4$, and $K = 5$) for each species and plotted results. Each vertical bar represents an individual, and color corresponds to the proportion of ancestry.

Figure S2: Length comparisons among each combination of genetic group and original sampling location for white sucker, smallmouth bass, and rock bass. BR = "Boardman River" group, GL = "Great Lakes" group, "up" refers to fish caught upstream of the Union Street Dam and "down" refers to fish caught downstream of the dam. Tukey's HSD was performed at $\alpha=0.05$ within each species, and different letters over box plots correspond to significantly different lengths.

Figure S3: Proportion of each putative genetic group within each sampling event for all five study species in the Boardman River. Numbers on bar plots represent sample sizes. BR = "Boardman River" group, GL = "Great Lakes" group.

CHAPTER 2: ENVIRONMENTAL DNA METABARCODING OUTPERFORMS TRADITIONAL FISHERIES SAMPLING AND REVEALS FINE-SCALE HETEROGENEITY IN A TEMPERATE FRESHWATER LAKE

INTRODUCTION

Assessing biodiversity is critical to understanding ecosystem function and informing conservation (Gotelli & Colwell, 2011; Iknayan et al., 2014), but estimating species richness can be challenging, especially in aquatic environments (Gu & Swihart, 2004). An accurate understanding of fish community composition and species diversity is nevertheless essential to making informed fisheries management and conservation decisions (Helfman, 2007). For decades, fisheries managers have implemented different standardized tools to sample fish communities (Bonar et al., 2009; Zale et al., 2013), often utilizing multiple gear types to account for known biases with individual methods (Ruetz et al., 2007; Schneider, 2000). For example, electrofishing, gill netting, trawling, and fyke netting all target different groups and sizes of fishes in different habitats and depths, and a combination of these techniques are often used for assessments (Bonar et al., 2009; Zale et al., 2013). However, even if multiple sampling methods are used, some gear selectivity will persist, and sampling may not accurately represent true community composition (Schneider, 2000; Zale et al., 2013). Additional challenges also exist with traditional sampling including misidentification of species in the field, high cost, significant infrastructure and labor requirements, potential destruction of habitats and organisms, and failure to detect rare or elusive species (Deiner et al., 2017; Gotelli & Colwell, 2011; Iknayan et al., 2014; Thomsen & Willerslev, 2015). Often, these rare species are of particular interest to managers (i.e. endangered or invasive), and therefore false absences could lead to erroneous

interpretations and inappropriate (or lack of) management action (Gu & Swihart, 2004; Thompson, 2013).

In recent years, environmental DNA (eDNA) metabarcoding has emerged as a useful tool for characterizing aquatic communities (Deiner et al., 2017; Thomsen & Willerslev, 2015). eDNA describes genetic material obtained from an environmental sample such as soil, sediment, snow, air, or water. As organisms interact with their surrounding environment, they shed DNA through excreted cells, sloughed-off tissue, gametes, and waste (Taberlet, 2012). This DNA can persist in the environment and be sampled to detect organisms without needing to physically handle specimens (Deiner et al., 2017; Rees et al., 2014). Metabarcoding utilizes universal primers that target an entire group of taxa in an eDNA sample, from which species barcodes are PCR amplified and sequenced on a high-throughput platform (Deiner et al., 2017; Porter & Hajibabaei, 2018).

Several studies have demonstrated that results from eDNA metabarcoding are comparable to traditional survey methods and/or long-term survey data to quantify fish community composition (e.g. Balasingham et al., 2018; Cilleros et al., 2018; Nakagawa et al., 2018; Pont et al., 2018). For example, Hanfling et al. (2016) collected water samples along established gill net sampling sites in a lake in the United Kingdom and detected 14 of the 16 fish species historically recorded there. Previous studies have also shown that that eDNA metabarcoding can detect more taxa than traditional survey methods in some instances (e.g. Afzali et al., 2020; Civade et al., 2016; Olds et al., 2016; Yamamoto et al., 2017; Zou et al., 2020). Additionally, Sard et al. (2019) demonstrated that eDNA metabarcoding could characterize 95% of a fish community with less sampling effort than traditional gear, and that eDNA detected aquatic invasive species that were not observed in some nets.

The field of eDNA metabarcoding has progressed substantially over the last decade, as researchers have developed reliable workflows for DNA extraction (Djurhuus et al., 2017; Lear et al., 2018), amplicon sequencing (Menning et al., 2018; Miya et al., 2015), and sequence analysis (Bolyen et al., 2019; Callahan et al., 2016a; Porter & Hajibabaei, 2018; Schloss et al., 2009). However, eDNA metabarcoding is still an evolving field and many improvements to sampling and study design could be warranted (Dickie, 2018; Ruppert et al., 2019). Some specific areas of eDNA metabarcoding studies that could be refined include sampling effort, amplicons/genes and databases used for taxonomic assignment, and the use of controls. For example, researchers may sample over a relatively large geographic area but collect only one, or very few, water sample replicates at a site or even for an entire lake. Additionally, some studies only amplify one gene region, which may lead to uncertainties in taxonomic assignment, or omit entire groups of taxa altogether (Shaw et al., 2016; Stat et al., 2017). Other studies lack proper use of negative controls in the field and/or in the laboratory. Finally, some studies compare eDNA to only one traditional gear type, which could lead to misleading results due to the problem of selectivity mentioned above.

Our goal was to intensively sample a small temperate lake over several weeks using two different traditional fish sampling methods and pair this sampling with eDNA metabarcoding of lake water samples using two mitochondrial DNA genes. Our first objective was to compare estimates of fish community composition between traditional and eDNA sampling methods. Our next objective was to assess how the different net types sample fish communities differently (e.g. species selectivity among gears) and to compare the taxa detected in lake water samples using the two metabarcoding genes (e.g. is a species detected with one gene but not the other?). Our

final objective was to determine if we could detect any temporal or spatial heterogeneity in fish community composition in the lake with either traditional gear or eDNA metabarcoding.

METHODS

Study location and field collection

The Boardman River watershed encompasses 740 km^2 in northwest lower Michigan (USA). In its lower reach, the river flows through Boardman Lake, a 137-hectare natural drowned river mouth lake, before meandering through Traverse City and then emptying into Grand Traverse Bay, an inlet of northeastern Lake Michigan (Figure 1). Boardman Lake and the upper Boardman River watershed are isolated from Lake Michigan by the Union Street Dam, an impassable earthen dam constructed downstream of Boardman Lake in 1867 roughly 1.5 km upstream of the river mouth (Kalish et al., 2018). Boardman Lake was selected for this study because the Great Lakes Fishery Commission (GLFC) is replacing the Union Street Dam with a fish passage structure to facilitate upstream movement of native fish species from Lake Michigan while blocking aquatic invasive species such as sea lamprey (*Petromyzon marinus*; Zielinski & Freiburger, 2020). While Boardman Lake is not a well-studied system, the GLFC is implementing several pre- and post-construction assessments in the lower Boardman River, including this metabarcoding research, to assess how fish assemblages and distributions may change in the future as a result of fish passage (http://www.glfc.org/fishpass.php).

We intensively sampled Boardman Lake in Traverse City, MI from May to June in 2019. The lake was split into 3 sampling zones, with Zone 1 being the upstream and southern end of the lake, Zone 2 in the middle, and Zone 3 being the downstream and northern end (Figure 1). We attempted to make these zones roughly equal in surface area, but the southernmost end of the

lake was extremely shallow and inaccessible to boats due to recent sediment deposition from dam removals upstream, so we omitted that portion of the lake from sampling and shifted zones northward. In each zone, we sampled 10 locations with five locations on the east side of the lake and five locations on the west side (Figure 1). We attempted to equally space sampling locations within each zone, but due to private property and the presence of docks, we occasionally had to shift sample sites. At each sample site, we set a mini fyke net against the shore and an experimental gill net 5 - 10 m out from the end of the fyke net, perpendicular to the shoreline. Each pairing of the two net types was later combined into a single "net group" for analysis, with a total of 60 individual nets set and 30 net groups in the lake. Fyke nets had rectangular frames that were 92 cm wide x 60 cm high, with two frames per net. Circular hoops were 60 cm in diameter with three per net, and the mesh size for the entire net was 9.5 mm. Net leads were 7.6 m in length and 92 cm high. Fyke nets were set so the rectangular frames were just beneath the surface of the water (0.7 – 1 m depth). Experimental gill nets were 38 m x 1.8 m with five 7.6 m graded meshes of 3.8 cm stretch, 5.1 cm stretch, 6.4 cm stretch, 7.6 cm stretch and 10 cm stretch. Gill nets were set with the narrow mesh towards the shore (shallow) and the larger mesh away from the shore (deep). Depth of gill nets varied among sites due to the variable nature of the lake's bathymetry. Depth of the shallow end of the net ranged from 2 – 4 m, and the deep end ranged from 3 - 12 m, depending on location. Fyke nets tend to sample shallow littoral species, and our mini fyke nets target smaller fish due to their relatively small mesh size. Gill nets tend to sample larger pelagic fish, and in general both nets rarely catch benthic or sedentary species (Zale et al., 2013).

At each net set location, we measured water temperature, recorded GPS coordinates, and assigned a unique waypoint number. From the boat and before setting the net, we collected three

1 L surface water samples into 1 L Nalgene wide-mouth HDPE bottles (ThermoFisher Scientific, Waltham, MA). Bottles were previously decontaminated by soaking in a 10% bleach solution for at least 10 minutes. Water samples were collected prior to setting nets to avoid any contamination from the net itself. Each collection event (per fyke or gill net) consisted of three replicates of lake water and one negative control ("field negative"), which was a 1 L Nalgene bottle filled with 1 L distilled water, brought into the field, and handled like all other samples. Gloves were always worn when collecting samples and were changed at each net set location. All bottles of water were stored on ice in a cooler that was decontaminated daily with 10% bleach. After collecting the three lake water samples (including one negative control), we set the corresponding gill or fyke net. We set one net group per zone each sampling day (three net groups, six total net sets) and left nets to fish overnight. The next day, we removed fish from nets, identified species, and recorded total length in mm for each fish. We performed 10 of these sampling days over a one-month period. With 60 individual net sets and three lake water samples and one negative control for each, we collected a total of 180 water samples and 60 negative controls from Boardman Lake.

In addition to the water samples corresponding to net sets in Boardman Lake, we also collected water samples below the Union Street Dam at two locations on the lower Boardman River. Since this river section is directly connected to Lake Michigan, we wanted to see if we could detect any species there that we did not observe in Boardman Lake. Water was collected from each location on three different dates from May to June 2019. As described above, each sampling event consisted of three replicates of river water plus one negative field control. We collected and filtered these water samples in the same manner as the Boardman Lake samples. With three sampling events at two sites consisting of three water samples plus one negative

control per site, we had a total of 24 water samples from the lower Boardman River. For later analyses, each downstream sampling location was designated as a unique "net group".

At the end of each sample collection day, we filtered 1 L of water from each bottle through 47 mm 0.45 µm nitrocellulose Nalgene analytical test filter funnels (Thermo Fisher) using a Gemini 2060 Dry Vacuum Pump (Welch, Prospect, IL). All filtering supplies and instruments were sterilized by soaking in a 10% bleach solution for at least 10 minutes. Between each sample, we changed gloves and sterilized the work benches. Negative controls were filtered in the same manner as lake water samples. Filters were preserved in 95% ethanol in individual 50 ml tubes and stored at room temperature. Including negative field controls and water samples from Boardman Lake and Boardman River below Union Street Dam, we processed a total of 264 water bottles (see Table S1 for water sample metadata).

Sequencing library preparation

Filters were cut in half; one half was used for extraction, and the other half was left in ethanol in its respective vial in the event we needed to re-extract filters due to potential problems downstream. Ethanol was allowed to evaporate from the half-filters for 24 hours before extraction. We followed the extraction methods outlined in Sard et al. (2019) and Laramie et al. (2015), using a combination of the Qiagen DNeasy Blood & Tissue Kit and the Purification of Total DNA from Animal Tissues Spin-Column Protocol (ThermoFisher). We performed two elutions of 100 µl each using Buffer AE warmed to 70°C. Eluted DNA was treated with Zymo OneStep PCR Inhibitor Removal columns (Zymo Research). Extractions were performed in a UV-sterilized laminar flow hood and all extraction instruments and bench spaces were sterilized with a 10% bleach solution or UV light. All pipetting was performed using sterile barrier filter tips.

Extracted DNA was then transferred from the elution tubes into 96-well plates, so that each plate contained 94 eDNA samples, a "lab negative" control (distilled sterile H_2O), and a positive control. Our positive control was DNA extracted from a caudal fin clip of *Pygocentrus natterei*, a tropical fish not found in our study system, which we included to visualize any potential cross contamination between wells. We then amplified extracted DNA at two mitochondrial genes, 12S and 16S, using primers also used by Sard et al. (2019) in their study on similar freshwater systems in Michigan. For 12S, the universal primers were originally developed by Riaz et al. (2011): Forward: 5′-ACTGGGATTAGATACCCC-3′, Reverse: 5′-TAGAACAGGCTCCTCTAG-3′. For 16S, we used universal primers originally developed by Deagle et al. (2009) and modified by Sard et al. (2019) to better match diversity in Michigan lakes: Forward: 5′-CGAGAAGACCCTNTGRAGCT-3′, Reverse: 5′CCKYGGTCGCCCCAAC-3′. Each core primer was synthesized with an Illumina tail on the 5' end complementary to the adapters used during barcoding described below. Forward primers were tailed with the Small RNA Sequencing Primer (5′-CGACAGGTTCAGAGTTCTACAGTCCGACGATC-3′), while reverse primers were tailed with the Multiplexing Read 2 Sequencing Primer (5′-GTGACTGGAGTTCAGACGTGTGCTCTTCCGATCT-3′).

PCR reactions were conducted in 10 µl volumes with 3 µl of DNA and 7 µl of a master mix. The master mix for the 12S gene contained 1X NEB 10X Standard Taq Reaction Buffer, and concentrations of 0.24 mM dNTPs, 2 mg/ml MgCl2, 1 mg/ml BSA, 1.25 U/µl NEB taq, and 0.8 µM forward and reverse primers. The master mix for the 16S gene contained 1X NEB 10X Standard Taq Reaction Buffer, and concentrations of 0.32 mM dNTPs, 2.5 mg/ml MgCl2, 1 mg/ml BSA, 1.25 U/µl NEB taq, and 0.8 µM forward and reverse primers. Thermal cycling was

performed as follows: 95 °C hold for 2 min, followed by 35 cycles of 95 °C for 30 s, 57 °C for 30 s for 12S and 45 s for 16S, 72 °C 45 s and a final extension of 72 °C for 5 min.

We then produced the final metabarcoding libraries using latter steps in the genotyping-in-thousands by sequencing (GT-seq) protocol (Campbell et al., 2015) following Bootsma et al. (2020). Briefly, each individual and plate was barcoded in a second indexing PCR, and barcoded products were normalized using SequalPrep DNA Normalization plates (Invitrogen, Carlsbad, CA). Normalized products were pooled within plates, and the pooled libraries were purified and size-selected using AMPure XP (Beckman Coulter, Brea, CA). Products were visualized on a 2% E-Gel EX agarose gel run on a Power Snap Electrophoresis Device (ThermoFisher) to verify size ranges and quantified using a Qubit 2.0 Fluorometer and dsDNA HS Assay Kit (ThermoFisher). Including positive and negative lab controls, we then sequenced 540 dual-indexed 12S and 16S amplicons on two Illumina MiSeq 2x150 flow cells at the University of Wisconsin Biotechnology Center DNA Sequencing Facility in Madison, WI.

Data filtering and quality control

We concatenated raw reads from both sequencing runs for each sample and used the program cutadapt (Martin, 2011) to remove adapter contamination from the 5' and 3' ends of the 12S and 16S sequences. We then processed cleaned data with the DADA2 package (Callahan et al., 2016a) in R (R Core Development Team). Due to variable sequence lengths, we retained reads between 130 and 145 bp for 12S, and 80 and 145 bp for 16S. For both genes we used the pool = TRUE parameter due to its increased sensitivity to rare sequences (Callahan et al., 2016b). After filtering, denoising, pooling, merging, and removing chimeras in DADA2, we exported sequence tables containing the final amplicon sequence variants (ASVs) and the number of reads per sample.

We used BLASTn (Altschul et al., 1990) to locally BLAST our 12S and 16S sequence tables to reference databases created by Sard et al. (2019), which included species common in Michigan lakes. Using customized R scripts, we parsed aligned ASVs and retained matches with greater than 98% sequence identity and alignment lengths greater than 140 bp for 12S and 80 bp for 16S. These lengths were chosen after exploration of raw data revealed robust 16S alignments down to 80 bp for taxa known to inhabit the Boardman River watershed but few robust alignments below 140 bp for 12S. After filtering, several ASVs still remained with no matches to our reference database, and we used BLASTn to align these sequences to the National Center for Biotechnology Information (NCBI) nucleotide database. We then used the same parameters as above to retain alignments. Sequences that were added through alignment to the NCBI database were combined with the original database to create a fully comprehensive reference. We then reBLASTed our sequences against the updated reference databases using BLASTn locally and ran outputs through our R script using the same cutoffs as above. Any ASVs with no matches after this step were removed, and all species other than fish were removed. ASVs with a single match that met our parameters or that had multiple matches for a single species (unambiguous) were assigned that species. ASVs with a match to two or more species (ambiguous) were either assigned back to common genus, or in some cases, to common family when multiple genera were present. However, some taxonomic groups presented exceptions to these rules due to the lack of within-genera variation within a particular gene and the potential for hybridization between closely related species. We thus collapsed any species in *Oncorhynchus*, *Lepomis*, *Rhinichthys*, and *Cottus* to genus. Finally, because multiple ASVs assigned to the same species, we collapsed all ASVs to unique assignments and summed reads for each sample.

To minimize false-positive detections, we accounted for contamination by subtracting reads in our controls from reads in water samples as follows: For each set of three water sample replicates, we subtracted the maximum number of reads for any single species found in the field negative, the lab negative, or the lab positive for each species other than *P. natterei*, (i.e., whichever control produced the highest per species contaminating read count) from all reads in the corresponding water samples. Species with more than 10 reads after subtracting contaminated reads were considered true hits. After accounting for contamination and removing hits with reads less than 10, we constructed community matrices for 12S and 16S using both read counts and presence/absence. We then grouped Boardman Lake samples by waypoint (three lake water samples) and net group (six lake water samples). For the samples taken below the Union St Dam, each waypoint was also considered a unique net group and consisted of nine samples (three sampling events consisting of three water samples each). A species was considered present at a given waypoint or net group if it was detected in at least one water sample replicate.

We also constructed presence/absence and abundance datasets for the traditional sampling gear to facilitate comparisons to the eDNA dataset. Abundance was the count of individuals from each species in each net, and species were considered present if they were found in the net. We then combined and summed abundances from each set of gill and fyke nets to acquire a total abundance of each species for each net group. To ensure that our comparison of species between methods was accurate, we changed the name of some taxa found in nets to match our eDNA assignments. For example, any *Lepomis macrochirus* detected in the eDNA was assigned to "*Lepomis* sp." and any species in the genus *Oncorhynchus* were assigned to "*Oncorhynchus* sp.", therefore, any *L. macrochirus* or *O. mykiss* species caught in nets were assigned back to genus for comparison with the eDNA data.

Statistical analyses

The first goal of our data analysis was to quantify and visualize basic trends in our data. We constructed boxplots of read counts and catches for each gene and gear and used Student's 2-sample t-tests for comparisons. We assessed the correlation between number of reads and number of unique taxa detected in each sample for both 12S and 16S using Spearman's correlation coefficient. We used Pearson's correlation to assess the relationship between the number of instances a taxon was detecting with 12S and 16S. Additionally, we constructed one heat map to visualize the species in each net group detected with 12S, 16S, or both genes, and a second heatmap to visualize the species detected with gill nets, fyke nets, or both gears. A species was considered "detected" if it appeared in any of the replicates for a given net group. We also constructed separate heatmaps for each gene quantifying the number of eDNA water samples (out of six) where a species was detected by each gene in each net group. Finally, we assessed trends in our negative control samples by constructing a boxplot of total reads in each negative control sample for each gene, a Pearson correlation comparing species-specific detections for 12S compared to 16S in negative controls, and a Spearman correlation comparing the number of eDNA detections for a given species to the number reads of that species found in negative controls summed across both genes. We set $\alpha=0.05$ for all tests.

To assess how each sampling method (12S eDNA, 16S eDNA, fyke nets, and gill nets) surveyed the fish community in Boardman Lake, we developed species accumulation curves using the "exact" method for the specaccum function in the R package vegan (Oksanen et al., 2019). Curves were constructed for all four methods separately and to compare eDNA (12S and 16S combined) with traditional sampling (fyke and gill nets combined). For combined eDNA

and combined traditional gear curves, a species was considered present if it was found in either gear or gene.

We correlated eDNA read counts to catch numbers and biomass of fish sampled in traditional gear to explore the relationship between eDNA read counts and fish abundance. This analysis was isolated to five species that were found most frequently in the traditional gear: *Ambloplites rupestris*, *Catostomus commersonii*, *Esox lucius*, *Sander vitreus*, and *Perca flavescens*. Pearson correlations (read counts in eDNA vs fish counts/biomass in nets) were conducted separately for each gene/gear combination for each of the five most common species. Biomass data for each net was estimated by converting total lengths to weights following species-specific conversion equations in Schneider et al. (2000).

A major objective of this project was to determine whether estimates of fish community composition were influenced by gear, gene, sampling method, spatial location, or other environmental variables. We used presence/absence data and two multivariate approaches, non-metric multidimensional scaling (NMDS) and redundancy analysis (RDA), to address this objective. NMDS was conducted to compare estimates of community composition between 12S and 16S, gill and fyke nets, eDNA (combined 12S/16S) and traditional gear (combined fyke and gill nets). NMDS was also used to compare community composition between samples taken in different zones, different sides of the lake (E vs W), and above and below the Union Street Dam. NMDS analysis by zone and side was conducted for both eDNA data (combined 12S/16S) and traditional data (combined gill and fyke net data), while NMDS comparing composition above and below the dam was only conducted for eDNA as no data for traditional gear were available below the dam. NMDS was conducted with the Bray-Curtis dissimilarity matrix and three ordination axes were generated for each analysis. We explored generating two and four

ordination axes and found that observed patterns were similar. We therefore chose to standardize our analyses to retain three ordination axes as this appeared to generally capture a large amount of the variation in the data while minimizing NMDS stress. NMDS analyses were visualized in R, and 95% confidence intervals around each group of datapoints were drawn with the dataEllipse function in the car package (Fox & Weisberg, 2019). We conducted analysis of similarities (ANOSIM; Clarke, 1993) for each NMDS dataset using the R package vegan to assess significant differences in community composition among analysis groups.

We conducted RDA analysis in the R package vegan for eDNA and traditional data to determine the influence of sampling location (zone and side of lake), water temperature, and sampling date on community composition. Our choice to conduct RDA instead of canonical correspondence analysis, a similar multivariate method, was informed by a detrended correspondence analysis, which indicated that variables were linear thus more appropriate for RDA (Legendre & Legendre, 1998). To ensure the robustness of our RDA analyses, we explored whether row normalizing our species presence/absence data or removing rare species (from one to three detections) influenced overall patterns. Additionally, we conducted RDA with all variables and with a single variable from each group of highly correlated variables as suggested by Hair (1995). The significance of each variable (i.e. their influence on variation in community composition) was assessed with ANOVAs. For both multivariate analyses, we used $\alpha=0.05$.

RESULTS

eDNA metabarcoding – 12S vs 16S

For 12S, we began with 12,702,289 raw reads across all samples. After filtering, merging, and chimera removal in the DADA2 pipeline, 7,249,412 reads remained (57.1%). Of

the 538 ASVs output from DADA2, 217 remained after applying our filtering parameters, aligning with reference sequences, and removing non-fish organisms. For 16S, we began with 13,202,450 raw reads across all samples. After the DADA2 pipeline, 3,458,683 reads remained (26.2%). Of the 8,855 ASVs output from DADA2, 784 remained after applying our filtering parameters and removing non-fish organisms. There were significantly more reads among samples in 12S (mean=25,407; SD=17,741) than 16S (mean=7,401; SD=10,739) ($p<0.001$, t = 13.522, df = 197) (Figure S1), but the average number of taxa detected by each gene did not significantly differ ($p=0.76$).

After combining filtered ASVs based on common assignments, both genes combined detected 40 unique taxa (Table 1, Figure 2). Both 12S and 16S individually detected 32 unique taxa, although taxonomic assignments were slightly different between the two genes (Table 1, Figures 2, S2, & S3). For example, 12S was able to unambiguously classify species in the genus *Etheostoma*, while 16S could not. 16S detected the *Lampetra* genus and *Petromyzon marinus*, while 12S had zero assignments to any lamprey species. Additionally, due to a lack of sequence variation in salmonids for the 12S gene, several ASVs assigned to more than one species or genus within the family, so we assigned these ASVs to "Salmonidae". For 16S, however, we did not need to assign back to Salmonidae because assignments were mostly unambiguous to genus. Both genes were often ambiguous for the Cyprinidae family, with a single ASV often assigning to multiple genera, so for both 12S and 16S we assigned these to "Cyprinidae". 16S appeared to be slightly better at classifying species in this family however, for example detecting *Luxilus cornutus* and *Pimephales notatus*. Overall, 12S detected four unique taxa, while 16S detected seven unique taxa (Table 1). The average number of taxa detected per net group were 14 (SD=3.95) and 13 (SD=3.5), for 12S and 16S, respectively (Table S2). There was a weak

positive correlation between the number of species detected in a sample and the number of reads for both 12S and 16S (Figure S4), and the number of detections of common species between genes was positively correlated (r = 0.91) (Figure S5). eDNA samples collected below the dam detected similar species to those in Boardman Lake, but some taxa were more common in, or exclusive to, the downstream samples, for example *Alosa sp*. (likely alewife), *Catostomus catostomus* (longnose sucker), *Coregonus artedi* (cisco), *Notemigonus crysoleucas* (golden shiner), *Percina caprodes* (common logperch), and *P. marinus* (sea lamprey) (Figures 2, S2, & S3). It is important to note that although some ASVs from water samples matched to *C. artedi*, it is possible that we could be detecting *C. clupeaformis* (lake whitefish), as this species is also present in Lake Michigan and very little sequence variation exists between these two closely related species.

Contamination in controls

Contamination in field negatives overall was low (i.e. generally less than 10 reads per species at maximum; Figure S6), although three field negatives contained high numbers (>1000) of reads. Two of these were contaminated with *S. vitreus* and one was contaminated with *C. commersonii* (Supplementary Files 1 & 2), two species that were common in the system based on net catches and reads in lake water samples. We removed these outliers for the analyses on field negative controls reported here and speculate on why they may have occurred in the discussion. There was a positive correlation between the number of detections for a species and the number of read counts in negative controls (Figure 3), suggesting that species more common in the system were also more likely to amplify in our field negative controls. Contamination by species was also correlated between 12S and 16S (r = 0.75; Figure S7), suggesting that contamination per species likely occurred before PCR or sequencing.

Our use of the lab positive control was successful, with both 12S and 16S amplifying tens of thousands of reads for *P. natterei* (Supplementary Files 1 & 2). Reads for this species in most other samples was zero, and if any reads existed, they were never greater than two, and were later removed during our filtering step. Contamination of other species in lab negative and lab positive controls was also low overall, with most species never amplifying, although there was a small amount of contamination from some species. For 12S, the highest number of reads in the lab controls was six (from *C. commersonii* in the lab positive control), and for 16S the highest number of reads in the lab controls was 13 (also from *C. commersonii* in the lab positive control) (Supplementary Files 1 & 2).

Traditional gear - Gill vs fyke nets

Using traditional gear, we caught a total of 12 species. Gill nets caught a total of 350 fish representing 8 unique species, and fyke nets caught a total of 195 fish representing 11 unique species (Table S2). Gill nets caught significantly more fish than fyke nets on average (mean=11.7, SD=7.6 for gill and mean=6.5, SD=4.5 for fyke, p=0.004; Figure S8). The average number of unique species caught in a single net were 4 (SD=1.431) and 2 (SD=1.048) for gill nets and fyke nets, respectively (Table S2). Species most common in gill nets were *P. flavescens* (38%), *E. lucius* (18%), *S. vitreus* (15.4%), and *A. rupestris* (14.6%). Species most common in fyke nets were *A. rupestris* (68.2%), followed by *P. flavescens* (9.7%) and *Neogobius melanostomus* (6.7%) (Table S3). Fyke nets caught 4 taxa that were not detected in gill nets (*Etheostoma sp*., *Lepomis sp.*, *Micropterus salmoides*, and *N. melanostomus*). Gill nets caught only one species that was not present in fyke nets (*S. vitreus*; Figure S9). In general, most species were either rare in nets or tended to be susceptible to only a single net type (Figure S9).

Traditional gear vs eDNA

Overall, eDNA metabarcoding detected significantly more taxa than traditional gear ($p<0.001$). Both 12S and 16S detected all 12 species caught using traditional gears (Table 1). These species were detected more frequently in the eDNA compared to traditional gears, or there were detections with both methods within a given net group (Figure 4). Only two net groups had a detection for a species observed in a net but not detected in the corresponding eDNA samples (one for *E. lucius* and the other for *S. vitreus*). Between 12S and 16S, 24 more unique taxa were detected in the eDNA than in the traditional gears (Table 1).

Among the 12 species detected using traditional gear, there appeared to be three main groups in terms of species abundance and catchability (Figure 5): (1) species present in low numbers in both nets and in eDNA (low abundance in the lake; e.g. *M. salmoides* and *Lepomis* sp.), (2) species present in low numbers in the nets but common in the eDNA (likely high abundance in the lake but with low catchability for our traditional sampling gears; e.g. *N. melanostomus* and *Etheostoma* sp.), and (3) species common in nets and in the eDNA (species very abundant in the system and also susceptible to our sampling gears; e.g. *A. rupestris* and *P. flavescens*).

The species accumulation curves demonstrate the higher number of species detected using eDNA compared to gill or fyke nets, and that relatively fewer eDNA samples are needed to reflect species richness in the lake (Figure 6). Also, gill nets detected more species than fyke nets at first, but with an increasing number of samples, fyke nets detected more species than gill nets overall (11 vs eight species, respectively). 12S and 16S had similar and mostly overlapping accumulation curves.

Overall, the number of reads in both 12S and 16S of our five most common species did not consistently correlate with either biomass or number of fish caught in each net, suggesting that read count data is unreliable to estimate abundance based on net catches, at least in this system (Figures S10 & S11). Of the 60 correlations we assessed, only one had a p value < 0.05, which compared the number of yellow perch caught in fyke nets to the number of yellow perch reads in the corresponding 12S eDNA water samples (Figure S10n).

Multivariate analysis results

Of the eight NMDS comparisons we performed to describe community compositions among variables and methods, there were five comparisons in which the ANOSIM analysis was significant: 12S vs 16S, Gill vs Fyke, eDNA vs traditional, eDNA by lake side, and eDNA above dam vs below dam (Table 2, Figures 7 & S12). For Figure 7, only the species with the top five loadings were included for ease of viewing, while Figure S12 contains all NMDS comparisons including all significant species. NMDS demonstrated clear differences between species caught with gill and fyke nets, with 95% confidence ellipses nearly completely separated (Figures 7a & S12a). The taxa with the highest loadings that were primarily driving these differences were *C. commersonii*, *E. lucius*, *M. salmoides*, *Etheostoma* species, and *A. rupestris*. Differences between 12S and 16S, although significant, were more subtle, and were not clearly driven by one or a few species (Figures 7b & S12b). Community composition estimates were highly differentiated between eDNA and traditional gear according to NMDS, with no overlap between 95% confidence ellipses and differences being driven by many species but primarily *Culaea inconstans*, *Cyprinus carpio*, Salmonidae, and *Salvelinus fontinalis* (Figures 7c & S12c). The NMDS of eDNA results grouped by lake side indicated significant spatial differences in community composition, with six net groups from the west side of the lake grouping closely

together and away from other net groups, that appeared to be associated with the presence of *Micropterus dolomieu* (Figure 7d & S12d). These net groups were 11, 17, 18, 22, 23, and 24 and were located across the entire west side of the lake (i.e. they were not all clustered in a single area of the lake). Taxa that differentiated samples from the east and west side of the lake included *M. salmoides, Umbra limi, Salmo trutta,* Cyprinidae, and *N. crysoleucas*. The two points below the dam clustered together and were slightly differentiated from other points above the dam in the NMDS based on eDNA results (Figure S12e). The species driving these differences included *S. trutta*, *C. artedi*, *Esox masquinongy*, and *P. caprodes*. The three other NMDS comparisons we performed (eDNA by sampling zone, traditional gear by lake side, and traditional gear by zone) were insignificant (Figure S12f-h).

RDA analysis demonstrated that no variables significantly influenced community composition estimated by traditional gear sampling (Table 3, Figure S13). For the eDNA sampling, however, all four variables (sampling date, zone, lake side, and water temperature) significantly influenced the community composition (Table 3, Figure 8). The most obvious differences in community composition existed between the six points on the west side of the lake (net groups 11, 17, 18, 22, 23, 24) identified in the NMDS analysis and all other points (Figure 8). The RDA analysis indicated that these points were differentiated by zone, lake side, and water temperature and identified similar species driving differences as were discussed above. Two of the environmental variables analyzed here showed high loadings on the same axes and are therefore correlated (lake side and water temperature) but removing one of these variables at a time still produced similar results.

DISCUSSION

Obtaining robust eDNA metabarcoding results

Using both 12S and 16S for our eDNA metabarcoding analysis provided a more comprehensive characterization of the fish community than if we had used a single gene (e.g. Evans et al., 2017; Sard et al., 2019; Shaw et al., 2016; Stat et al., 2017). Both genes consistently detected the most common species in the system, but rarer species were not always detected with both genes in a given sample. For example, in some net groups *M. dolomieu* (smallmouth bass) was detected just with 12S, in others it was detected only with 16S, and sometimes it was detected with both genes. Other rare species showed similar patterns, but no species that were resolved with both genes consistently amplified at one gene and not the other, indicating that variation in detections is likely a result of subsampling of extracted DNA or stochasticity in PCR rather than inherent differences in species-specific detection probabilities (Deiner et al., 2017; Kebschull & Zador, 2015). Taxonomic resolution varied between the two genes, with each classifying various taxonomic groups differently. For example, 12S contained sufficient variation to classify species within the *Etheostoma* genus (darters) while 16S could not. It is important to note that because of the difficulty of parsing some taxa to species, we sometimes assigned back to genus or family, and this could underestimate the total number of species present in the lake. For example, there are likely several species present within our Cyprinidae assignment for 12S, but we were not confident in those species assignments due to low sequence variation for that particular gene within that taxonomic group. Therefore, if a study requires high taxonomic resolution for a particular taxonomic group like Salmonidae or Cyprinidae, it would be beneficial to utilize additional genes that contain sufficient variation among taxa.

Our study demonstrates the importance of taking many water samples across time and space, especially to detect rare species. For example, *Pomoxis nigromaculatus* and *L. cornutus* were detected in water samples from only two out of 32 net groups, and *P. notatus* and *P.*

caprodes were only detected in one. Taking samples in triplicate was also essential. We sometimes observed that even common species were only detected in one replicate out of three, meaning if we had only collected a single water sample, that species likely would have gone undetected. It is well known that PCR bias and random events from sample collection to bioinformatics can lead to false negatives, even when extreme care is taken (Goldberg et al., 2016; Porter & Hajibabaei, 2018; Thomsen & Willerslev, 2015). This emphasizes the importance of using multiple water sample replicates per sample site, which has also been suggested by other studies (e.g. Evans et al., 2017; Shaw et al., 2016; Willoughby et al., 2016), but has not consistently been implemented (Dickie, 2018). We show that having many samples with replicates allows much higher taxonomic coverage, increased likelihood of detecting rare species, and overall high confidence in results.

Our metabarcoding data provided some useful information beyond our main study objectives. For example, we were able to compare species detected below the Union Street Dam to those in Boardman Lake. Community compositions were relatively similar between locations, but several species appeared to be more common below the dam, including *C. artedi* (cisco or lake whitefish), *P. marinus* (sea lamprey), and *Alosa sp.* (likely alewife). It is logical that these species common in the Great Lakes would be present in the lower river section connected to Lake Michigan. We also explored the potential of detecting DNA transported downstream from the upper Boardman River into Boardman Lake. It is well-documented that rivers transport eDNA downstream (e.g. Civade et al., 2016; Deiner et al., 2016; Jane et al., 2015; Pont et al., 2018); however, we did not detect any spatial patterns that would suggest a higher concentration of DNA from lotic species (e.g. brook trout *S. fontinalis*) where the river enters the lake. Based on the consistent distribution of eDNA around the lake, it is likely that most of the DNA we

detected in water samples were from fish residing in the lake. However, the ecology and decomposition rates of eDNA in this system are unknown and we had no upstream water samples to compare to; therefore, we cannot make concrete conclusions about the origin of fish DNA in the lake. Finally, while this research was not focused on non-fish species, we were intrigued to discover that several other animals amplified in our eDNA water samples. These included a variety of birds (duck, goose, swan, cormorant, chicken), mammals (squirrel, mole, raccoon, beaver, red fox, and black bear), and snapping turtle. These non-target detections provide useful information on community composition that could be mined in future studies.

Quality control with eDNA metabarcoding

Our study emphasizes the importance of meticulous consideration of all taxonomic assignments, knowledge of species phylogenies, and understanding of species distributions and potential for hybridization. We developed a fully comprehensive and curated reference database and passed our ASVs through conservative filtering parameters. Some ASVs matched to species known not to be present in the region, for example *Sander lucioperca* (zander) which exists in Europe and Asia but is related to native species *S. vitreus* and *S. canadensis*. These foreign taxa known to not be in the region were removed to avoid any impossible matches and create a more refined reference database. After filtering, rather than simply accepting the top match, each retained ASV assignment was assessed individually before being accepted as valid. Some assignments were straightforward; for example, one ASV had three matches, all of which were variants of *C. commersonii*. However, one ASV had a 100% match to *Ameiurus melas*, a 99.3% match to *Ameiurus nebulosus*, and a 98.6% match to *Ameiurus natalis*. It would be easy for a researcher to observe the first 100% match and promptly assign this ASV to *A. melas*, but because there was also high similarity to two other species, we were not confident that this ASV

was definitively *A. melas*. Therefore, we assigned that ASV to "*Ameiurus* sp." We also made these more conservative taxonomic assignments in instances where there is known low sequence variation within a group (e.g. Salmonidae with 12S and *Cottus* with both genes), or when there was a potential for species hybridization (e.g. in the genus *Lepomis*; Avise & Saunders, 1984). As mentioned above, these broader assignments may be undercounting true species richness within the system, but we felt it was important be completely confident in each taxonomic assignment.

In addition to ensuring confidence in taxonomic assignments, we chose to be quite conservative when determining if a species was truly present in a sample. Because we commonly detected hundreds or thousands of sequence reads for each species in our water samples, we set our minimum read requirement as 10. This is more conservative than other metabarcoding studies, which have set their minimum required reads as 2 or 3 (Balasingham et al., 2018; Olds et al., 2016). There is clearly the potential to lose some true detections with our higher cutoff, but once again we felt it was important to be conservative and confident in our detections. The fact that we observed 40 taxa even with this conservative data filtering suggests that our metabarcoding assay was sensitive and our methods were robust.

Most of our field negative controls had low contamination, proving that our sterilization techniques were effective. In instances where contamination did exist, it was often from one of the most common species in the Boardman system according to net sampling, suggesting that the contamination likely occurred during sample collection or filtering rather than during eDNA extractions or pre- and post-PCR. The low contamination in our laboratory positive and negative controls also verify that contamination from the laboratory, as well as cross contamination among plates and wells, was minimal. Surprisingly few studies report sources of contamination

or when it occurred during the process, but Harper et al. (2019) found that most contamination occurred during sampling and PCR, and Furlan et al. (2020) found that some contamination occurred through all stages of sampling and processing, but mostly during PCR. Because we subtracted the maximum number of reads in controls from all read counts in associated samples, we were confident that the remaining reads were true detections. Our results emphasize the importance of using field and lab controls to track any potential contamination, and accounting for any contamination during downstream bioinformatics.

Comparing eDNA metabarcoding to traditional surveys

In Boardman Lake, more species were detected with 12S and 16S eDNA metabarcoding than traditional survey gears in all net groups. Our eDNA methods detected all the fish species captured in traditional gears, as well as 24 additional taxa. Several species appeared often in the eDNA water samples but were observed in few to none of the nets. For example, *Rhinichthys* sp. (daces), *Lampetra* sp. (lampreys), *Percopsis omiscomaycus* (trout perch), and *Cottus* sp. (sculpins) were present in water samples from over half of the net groups in Boardman Lake but were never observed in nets. *N. melanostomus* (round goby) and *Etheostoma* sp. (darters) were captured in a low number of fyke net sets (5 and 7, out of 30, respectively), but *N. melanostomus* was present in water samples from every net group, and *Etheostoma* sp. were present in 29 out of 30 net groups. Our metabarcoding data suggest that these species are quite common in the lake but are not susceptible to sampling by either of our traditional gears. A common characteristic of these fishes is they are generally sedentary, small-bodied, and/or occupy benthic habitats (Becker, 1983; Scott & Crossman, 1985). These qualities make them difficult to capture with traditional gears, and the species selectivity of these sampling methods is well-established (Zale et al., 2013). Our study thus demonstrates that eDNA metabarcoding can overcome problems

with selectivity and the difficulty of capturing rare, small, and/or benthic fish species with traditional gears.

We also observed a clear difference in the species selectivity between our two traditional gear types. For example, *A. rupestris* was the most common species in fyke nets (68.2% of all fish), followed by *P. flavescens* (9.7% of all fish). In gill nets, however, *A. rupestris* only accounted for 14.6% of all fish caught. Our species accumulation curve demonstrated that at a low quantity of samples, gill nets detect more species than fyke nets, but with increasing samples, fyke nets detected more species overall. This could be because gill nets tended to catch more fish on average but they were larger, more pelagic fishes. Our fyke nets had a smaller mesh size and were set against the shoreline; therefore, they may be more likely to capture smaller, littoral, and/or benthic species. For example, we observed *Etheostoma* sp. and *N. melanostomus* in fyke nets but never in gill nets. These results further demonstrate that multiple gear types are necessary for accurate fish community assessments using traditional gear and highlight the advantage of lower species selectivity with eDNA.

Overall, the number of sequence reads and fish biomass/catch in Boardman Lake was not correlated. Several studies have demonstrated that read counts can correlate to fish abundance (e.g. Hanfling et al., 2016; Lacoursière-Roussel et al., 2016; Thomsen et al., 2016), however many of these studies obtained accurate population estimates using standardized sampling methods and/or long-term data. These data were unavailable for Boardman Lake, and we hypothesized that the number of fish encountered in a net should be relatively correlated to their local abundance, which may be incorrect. It is also possible that the lack of correlation could be due to error introduced during the eDNA sample collection and laboratory processing, or even the introduction of genetic material from upstream. In addition, subsampling, primer bias, and

PCR inhibition can lead to a lack of correlation between eDNA sequence reads and true abundance (Deiner et al., 2017; Goldberg et al., 2016; Porter & Hajibabaei, 2018), although we attempted to pre-emptively address these issues through replication, use of vetted and published assays, and inhibitor removal. Additionally, ecological factors such as high numbers of gametes in the water during spawning season could impact correlations. Both *P. flavescens* and *C. commersonii* were common in the system, and if they had recently spawned, amplification of eDNA from their gametes could have inflated read numbers. In general, our data suggest that the proportion of detections across all eDNA water samples may be a better predictor of relative abundance than read counts, as higher occurrences in samples are likely a result of species being common in the system. For example, *P. flavescens* was caught with traditional gear in 29 out of 30 net groups and was present in water samples in 30 out of 30 net groups, suggesting it is likely one of the most common species in Boardman Lake.

Detection of spatial heterogeneity of fish distribution with eDNA metabarcoding

We were able to detect differences in fish community composition between the west and east sides of Boardman Lake with multivariate analyses of our eDNA metabarcoding data, but not with the traditional sampling data. This result suggests that our eDNA methods were sensitive enough to detect some spatial heterogeneity in fish distributions across relatively short distances (500 meters between the east and west shores). There were six net groups from the west side of the lake that appeared to be driving the differences we observed. Based on our NMDS comparison of eDNA data by lake side, *M. dolomieu* (smallmouth bass) appears to be the main species causing these six points on the west side of the lake to differ from the others. It is well-known that fish have inherently patchy distributions due to spatial differences in habitats as well as variations in both abiotic and biotic factors (Romare et al., 2003; Smokorowski & Pratt,

2007; Weaver et al., 1997). Therefore, our results suggest that there is likely some physical or biological characteristic of the habitat on the west side of the lake that *M. dolomieu* prefer. This species is known to congregate in areas with dense underwater woody debris (Becker, 1983; Scott & Crossman, 1985), but this was not a variable that we assessed as part of this study. Lake bathymetry was also not a variable that we explicitly measured, but in the field, we observed that the lake bottom on the west side dropped off more abruptly. This physical attribute may have potentially affected the presence or distribution of fish in those locations.

Because of the inherent patchiness of fish distributions within a system (Brind'Amour et al., 2011; Brind'Amour et al., 2005), it is reasonable that with enough samples, eDNA metabarcoding could detect some spatial differences. This further emphasizes the importance of collecting numerous eDNA samples across multiple locations to fully characterize the fish community within a lake. However, the analysis we conducted was post-hoc and our study was not designed to test hypotheses about which environmental variables might be correlated with fish distributions. To test such hypotheses, future metabarcoding studies could incorporate detailed habitat surveys and pair them with intensive water sampling. For example, researchers could include water quality measurements like DO, temperature, and turbidity; physical variables like substrate composition, bathymetry, cover, and woody debris; and biological variables such as aquatic vegetation and zooplankton abundance/composition.

Guidance for future studies and conclusions

In conclusion, we demonstrated that surface water eDNA detected roughly three times as many unique fish taxa than traditional methods in a small temperate lake. Our eDNA metabarcoding assay characterized the fish community much more comprehensively than traditional net sampling, providing evidence that metabarcoding can overcome some of the

problems associated with traditional sampling and provide a more sensitive approach to describe community composition. eDNA sample collection is fast and simple, and laboratory and analytical processes are becoming more streamlined. We therefore believe that eDNA is the clear choice to estimate fish species richness, as the personnel and infrastructure required to collect eDNA samples is minimal compared to traditional techniques. This decrease in cost will allow researchers and managers to sample more areas, sample more frequently, and conduct more intensive sampling, which will provide extremely valuable information to inform management and conservation of aquatic systems. However, eDNA metabarcoding is not expected to replace standard fisheries techniques for many applications such as studies on population dynamics, as eDNA does not distinguish between live or dead organisms, nor provide information on age, sex, or sizes of fish. Therefore, eDNA metabarcoding is likely best used to complement traditional fisheries methods, rather than replace them. Our results reveal the utility of collecting many samples across time and space, and the necessity of collecting multiple replicates for each sampling event. Additionally, we demonstrated that fine-scale heterogeneity can be detected using eDNA. Fish community structure is not homogenous across water bodies and collecting spatially representative samples is critical. Finally, this study demonstrates the importance of understanding species phylogenies and taking a conservative approach when classifying taxa and assigning species to achieve robust results.

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Table 1: Occurrences of all detected taxa using eDNA (12S and 16S genes) and traditional sampling gear (gill nets and fyke nets). Values in "12S" and "16S" columns are the number of net groups out of 32 in which a particular taxon was detected, and values in the "fyke" and "gill" columns are the number of net groups out of 30 in which a species was detected. Darker backgrounds indicate higher values. Asterisks indicate broader taxa (genus or family) in which a narrower assignment occurred for that method, which we used to avoid double-counting redundant assignments (for example, *Etheostoma* sp. for the 12S gene which was able to detect both *Etheostoma exile* and *Etheostoma nigrum*), therefore only taxa without associated asterisks were counted as "unique" for each method.

Scientific Name	Common Name	12S	16S	Fyke	Gill
Alosa sp.	River herrings	2	1		
Ambloplites rupestris	Rock bass	28	29	28	15
Ameiurus sp.	Bullheads	3	1		
Catostomidae	Suckers	1	*	*	*
Catostomus catostomus	Longnose sucker	2	3		
Catostomus commersonii	White sucker	32	32	5	18
Coregonus artedi	Cisco	3			
Cottus sp.	Sculpins	28	26		
Culaea inconstans	Brook stickleback	10	11		
Cyprinidae	Minnows & carps	18	14		
Cyprinus carpio	Common carp	10	6		
Esox lucius	Northern pike	31	28	3	22
Esox masquinongy	Muskellunge	1			
Esox sp.	Pikes	3	*	*	*
Etheostoma exile	Iowa darter	25			
Etheostoma nigrum	Johnny darter	13			
Etheostoma sp.	Darters	*	30	5	
Lampetra sp.	Lampreys		19		
Lepomis sp.	Sunfishes	8	7	2	
Luxilus cornutus	Common shiner		2		
Micropterus dolomieu	Smallmouth bass	14	20	2	1
Micropterus salmoides	Largemouth bass	5	4	1	
Neogobius melanostomus	Round goby	32	31	7	
Notemigonus crysoleucas	Golden shiner	1	2		
Oncorhynchus sp.	Pacific salmon & trout	24	24	1	1
Perca flavescens	Yellow perch	32	31	7	27
Percina caprodes	Common logperch		1		
Percopsis omiscomaycus	Trout perch	24	22		
Petromyzon marinus	Sea lamprey		2		
Pimephales notatus	Bluntnose minnow		1		
Pomoxis nigromaculatus	Black crappie	1	2		
Rhinichthys sp.	Riffle daces	13	18		
Salmo trutta	Brown trout	22	20	1	3
Salmonidae	Salmonids	15	*	*	*
Salvelinus fontinalis	Brook trout		4		
Salvelinus namaycush	Lake trout		1		
Salvelinus sp.	chars and trouts	4	*		
Sander vitreus	Walleye	28	24		20
Semotilus atromaculatus	Creek chub	8	3		
Umbra limi	Central mudminnow	4	3		
	n unique	**4**	**7**	**0**	**0**

Table 2: Results from analysis of similarities (ANOSIM) comparing estimates of community composition among different sampling methods and areas. P-values <0.05 are in bold. See NMDS plots in Figs. 7 and S12 for visualizations of community compositions.

Comparison	Anosim R	P-value
12S vs 16S	0.05	**0.002**
gill vs fyke	0.67	**0.001**
eDNA vs traditional	0.94	**0.001**
eDNA by zone	0.07	0.053
traditional by zone	0.02	0.238
eDNA by lake side	0.11	**0.010**
traditional by lake side	0.03	0.755
eDNA above vs below dam	0.61	**0.007**

Table 3: Results from redundancy analysis (RDA) to assess which variables significantly influenced estimates of community composition for eDNA and traditional sampling methods using presence/absence data. Variables assessed were sampling date, zone, lake side, and water temperature. ANOVA P-values <0.05 are in bold.

Sampling method	Variable	Variance	F-statistic	P-value
eDNA	Sampling date	0.22	1.73	**0.036**
eDNA	Zone	0.30	2.39	**0.004**
eDNA	Lake side	0.30	2.42	**0.003**
eDNA	Water temperature	0.28	2.25	**0.006**
Traditional	Sampling date	0.05	1.15	0.310
Traditional	Zone	0.06	1.39	0.176
Traditional	Lake side	0.02	0.50	0.882
Traditional	Water temperature	0.07	1.52	0.150

Figure 1: Sampling area with locations of net groups. We divided Boardman Lake into 3 sampling zones, each of which contained 10 net groups. Each net group consisted of one fyke net and one gill net and their corresponding lake water samples (3 per net). Water samples were also collected at 2 locations below the Union Street Dam on the lower Boardman River, but no corresponding nets were set because that river section is narrow and flowing with boat/kayak traffic

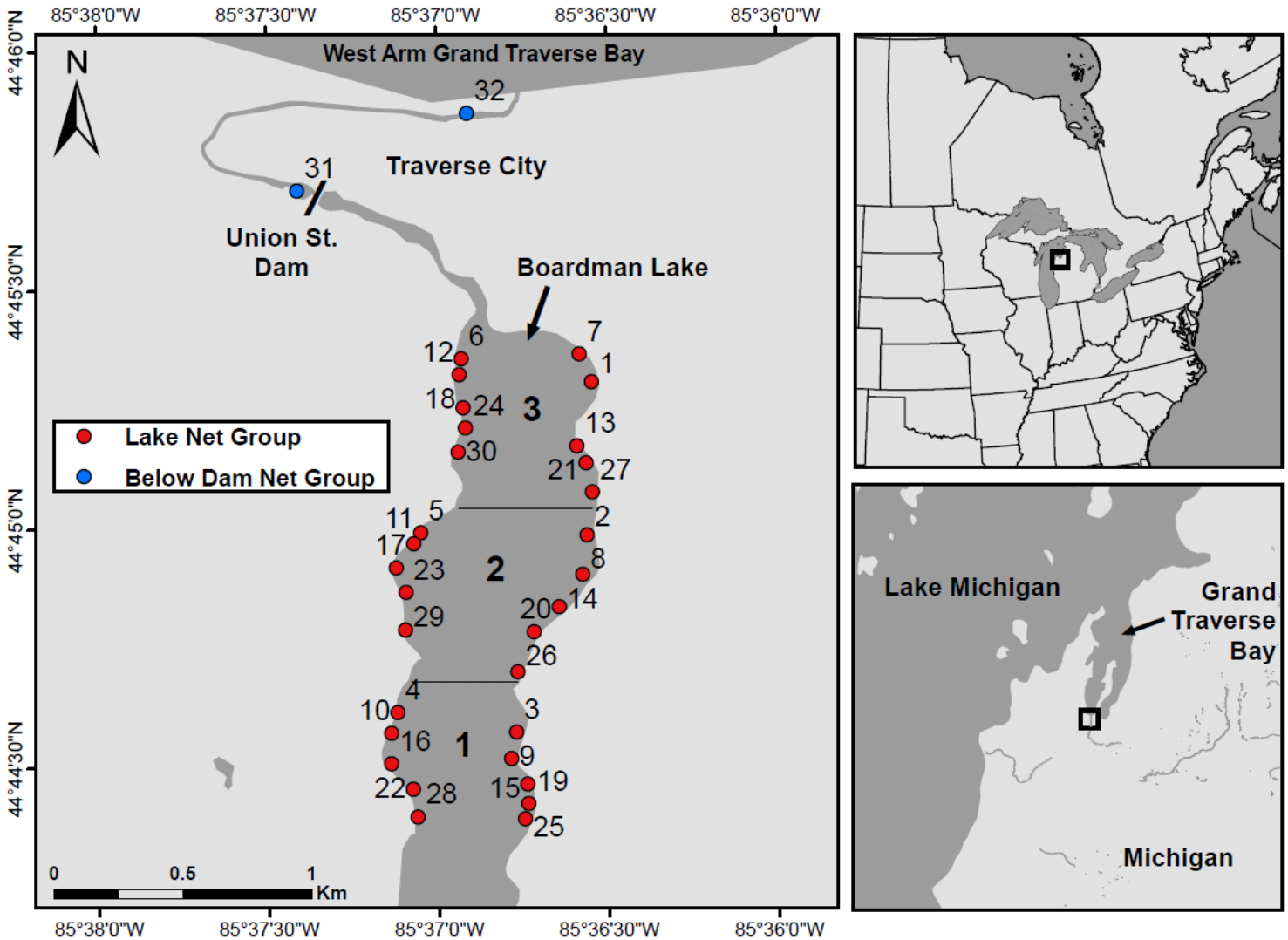


Figure 2: Heatmap indicating presence or absence of all potential taxa by net group for each gene used for eDNA metabarcoding. Blue indicates taxa present in a net group that was detected only with 12S, red indicates a detection only with 16S, purple indicates that taxa were detected with both genes, and grey signifies no detection. Each net group consisted of one fyke net and one gill net and their corresponding lake water samples (3 per net). Note that net groups 31 and 32 are the below-dam river sample sites.

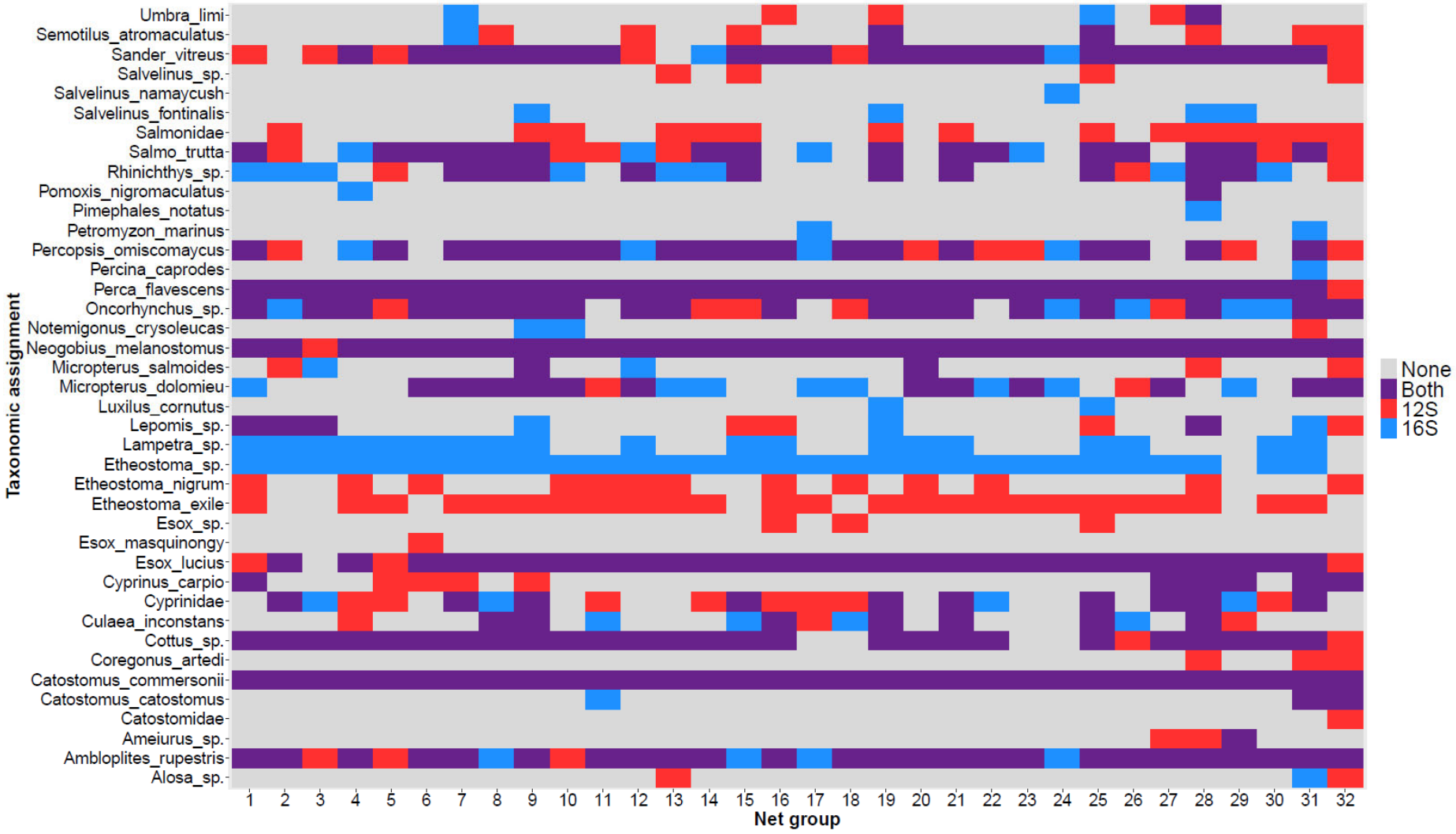


Figure 3: Relationship between the number of detections across all eDNA samples and number of reads in all field negative controls for all detected taxa assessed with Spearman's correlation coefficient (r_s=0.935). Each point represents a unique taxon. Three outlier negative controls with high read counts were removed prior to analysis (see results). Data includes both Boardman Lake and below-dam samples.

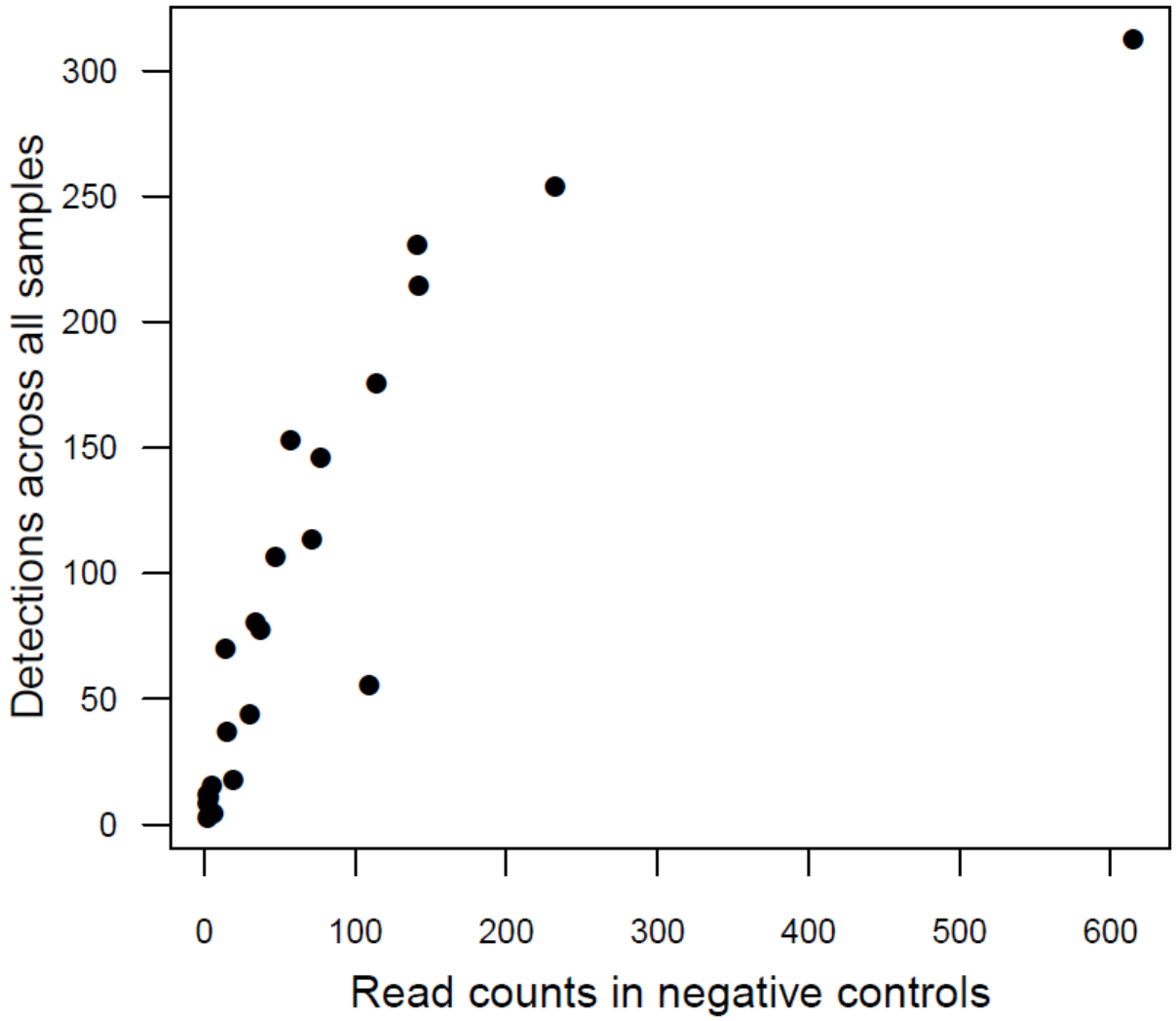


Figure 4: Heatmap indicating presence or absence of all species detected using traditional sampling gears (both fyke and gill nets), and comparison with corresponding detections using eDNA (12S and 16S combined) for each net group in Boardman Lake. Blue indicates that a particular species was detected in a given net group with eDNA only, red indicates it was detected only using traditional sampling, purple indicates it was detected using both methods, and grey signifies no detection with either method.

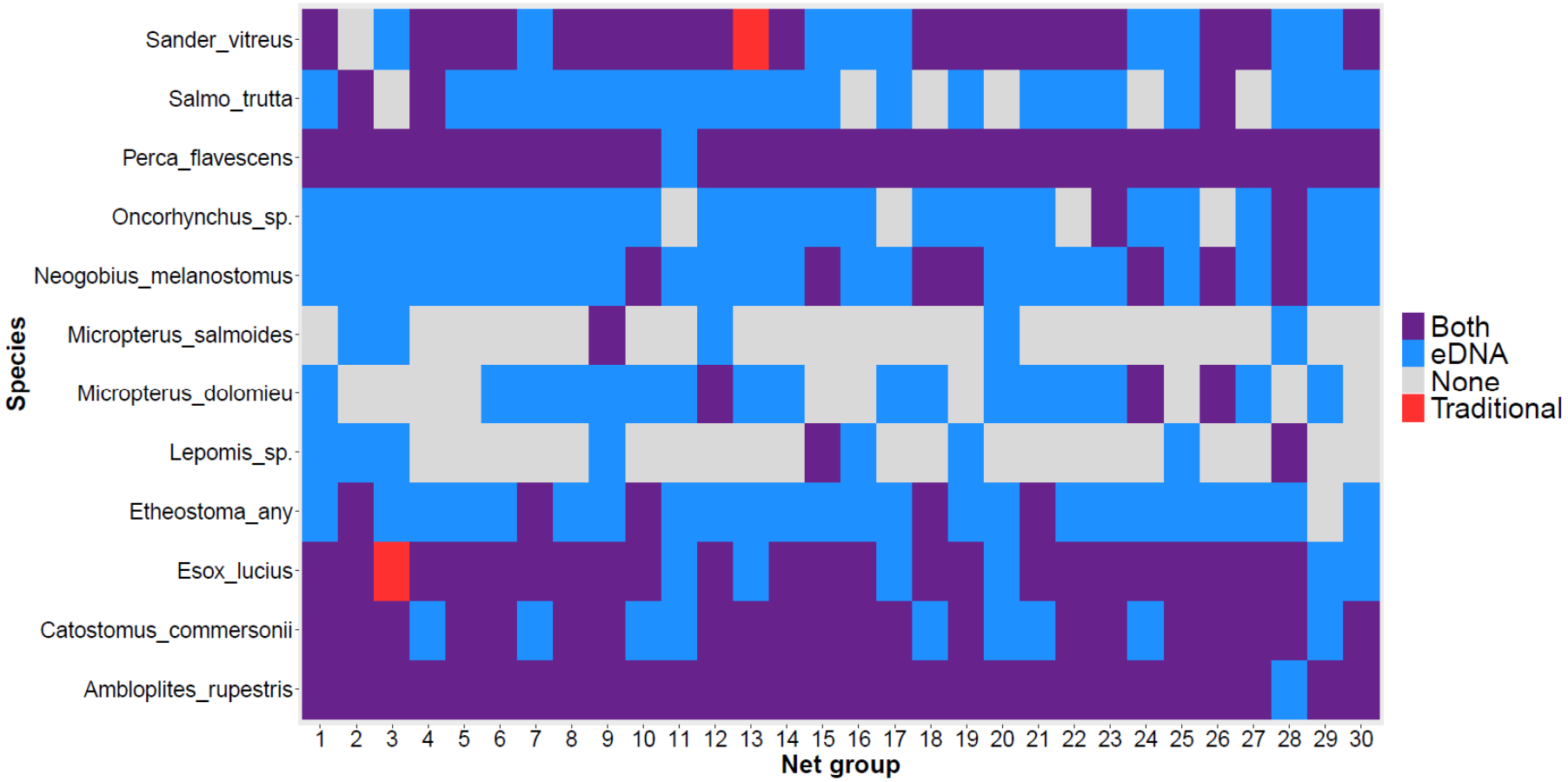


Figure 5: Relationship between the number of detections in eDNA (12S and 16S combined) and detections using traditional gear (fyke and gill nets combined) for species that were detected using both methods in Boardman Lake, with colors representing 3 distinct detection groups. Axes represent the number of net groups in which a taxon was detected. Red represents species that had low presence in both eDNA and traditional gears, suggesting low abundance in the system. Blue indicates species that were commonly detected using eDNA but not traditional gear, suggesting they are abundant in the lake but not susceptible to net sampling. Purple represents species with high detection rates using both methods, suggesting they are abundant in the lake and also susceptible to traditional sampling gears.

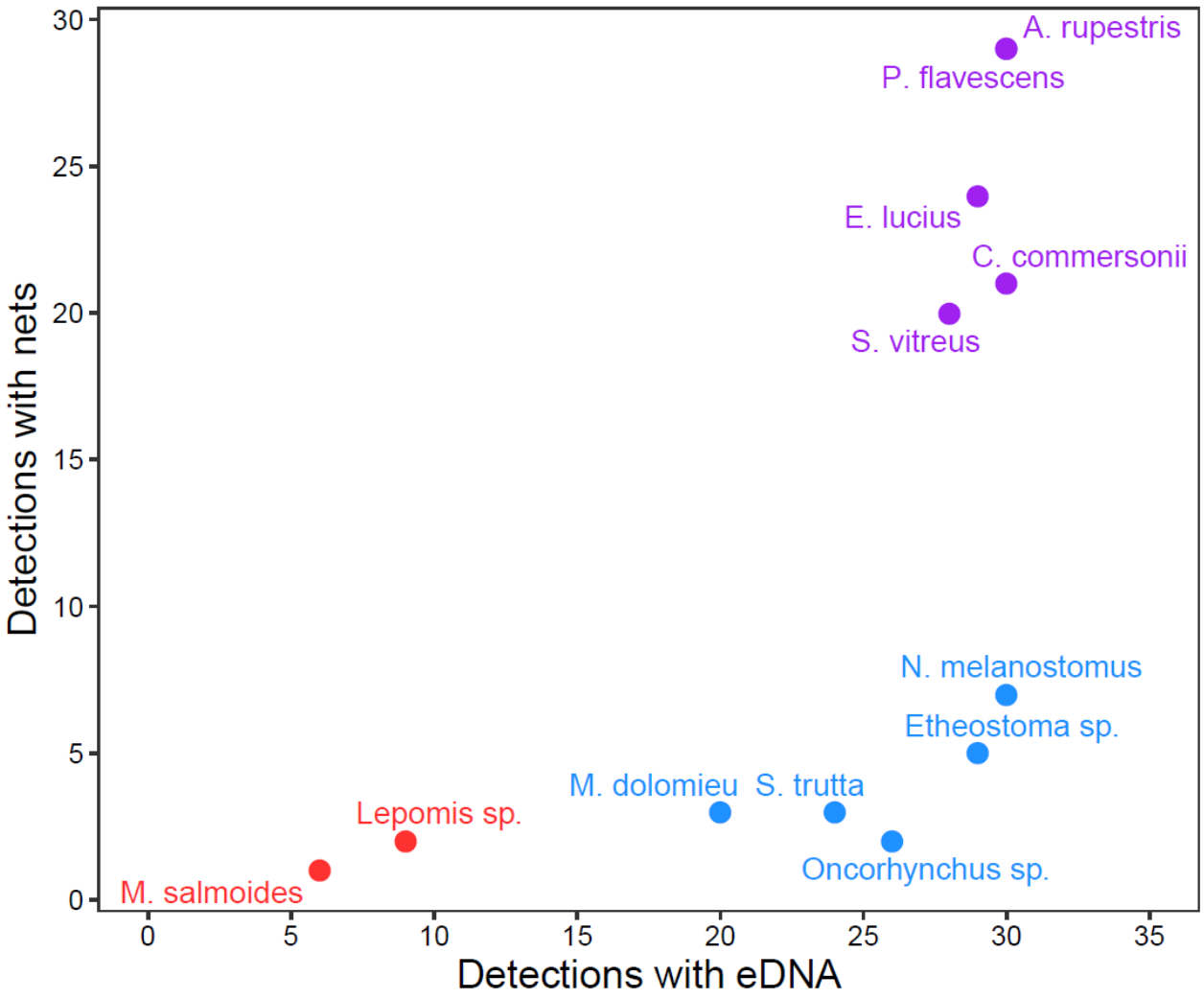


Figure 6: Species accumulation curves for Boardman Lake for a) each detection method based on species presence at individual waypoints (i.e. a net site where a fyke or gillnet was set and an eDNA sample was taken), and b) eDNA (12S and 16S combined) compared to traditional gear (gill and fyke nets combined) based on species presence in net groups. Grey shading represents the standard deviation of the expected number of species per sampling method.

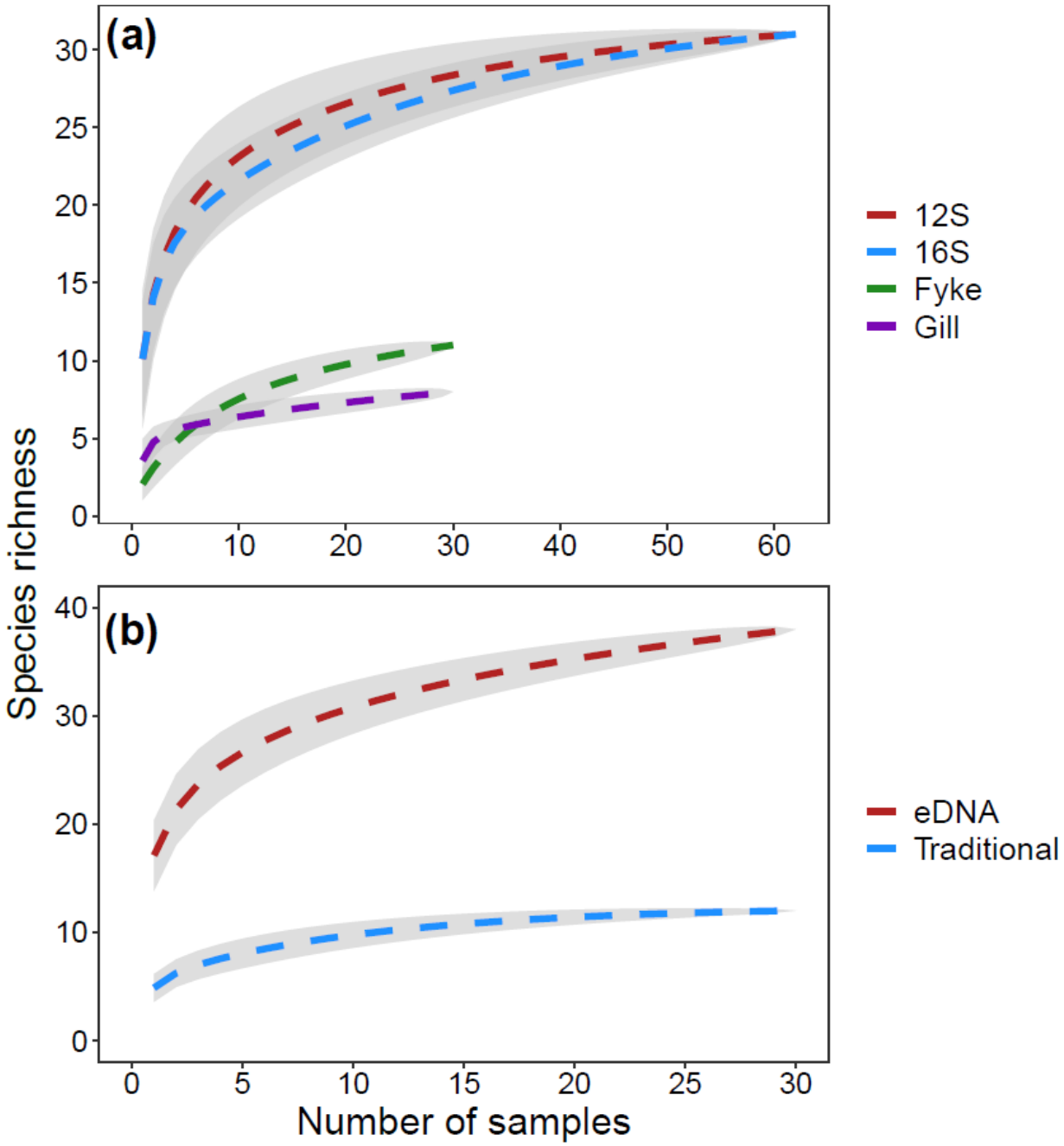


Figure 7: Non-metric multidimensional (NMDS) plots including 95% confidence interval ellipses for four of five significant comparisons (ANOSIM $p<0.05$). Each point represents a net group. For each plot, only the five species with the top loadings were included to facilitate visualizations. 3-letter codes represent the following taxa: A.RU = *Ambloplites rupestris*, C.AR = *Coregonus artedi*, C.CA = *Cyprinus carpio*, C.CO = *Catostomus commersonii*, C.IN = *Culaea inconstans*, CYP = Cyprinidae, E.LU = *Esox lucius*, ETH = *Etheostoma* sp., M.SA = *Micropterus salmoides*, N.CR = *Notemigonus crysoleucas*, ONC = *Oncorhynchus* sp., P.CA = *Percina caprodes*, SAL = Salmonidae, S.FO = *Salvelinus fontinalis*, S.TR = *Salmo trutta*, U.LI = *Umbra limi*. Plot (a) compares fyke nets and gill nets (traditional gear), (b) compares the 12S and 16S genes used in eDNA metabarcoding, (c) compares eDNA metabarcoding with traditional gear overall, and (d) compares the east and west side of Boardman lake from the eDNA data. The only other significant comparison not included here is the comparison between samples taken above and below the dam (S Fig 12e). See supplementary figure 12a-h for all NMDS plots including all significant species loadings.

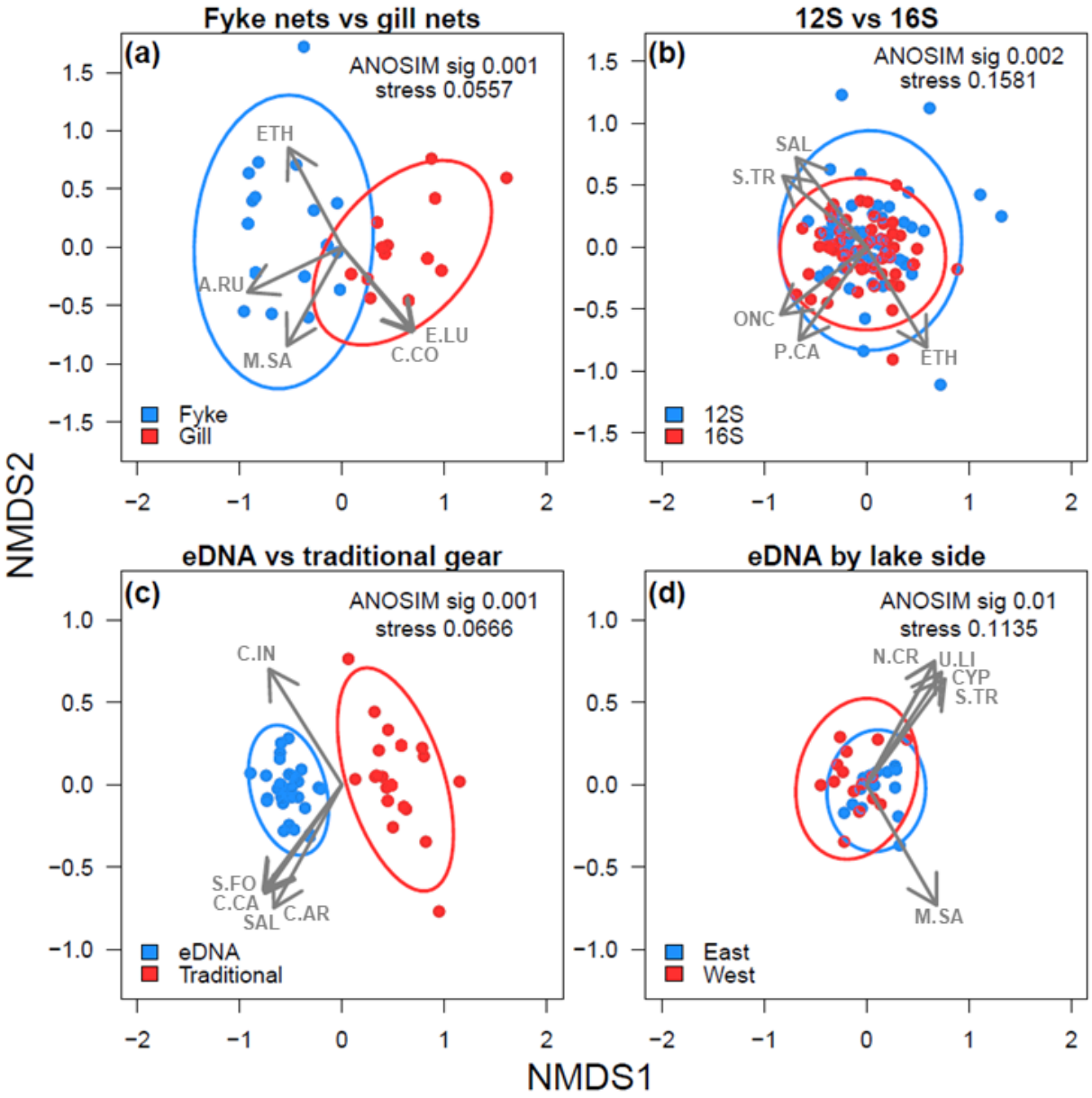
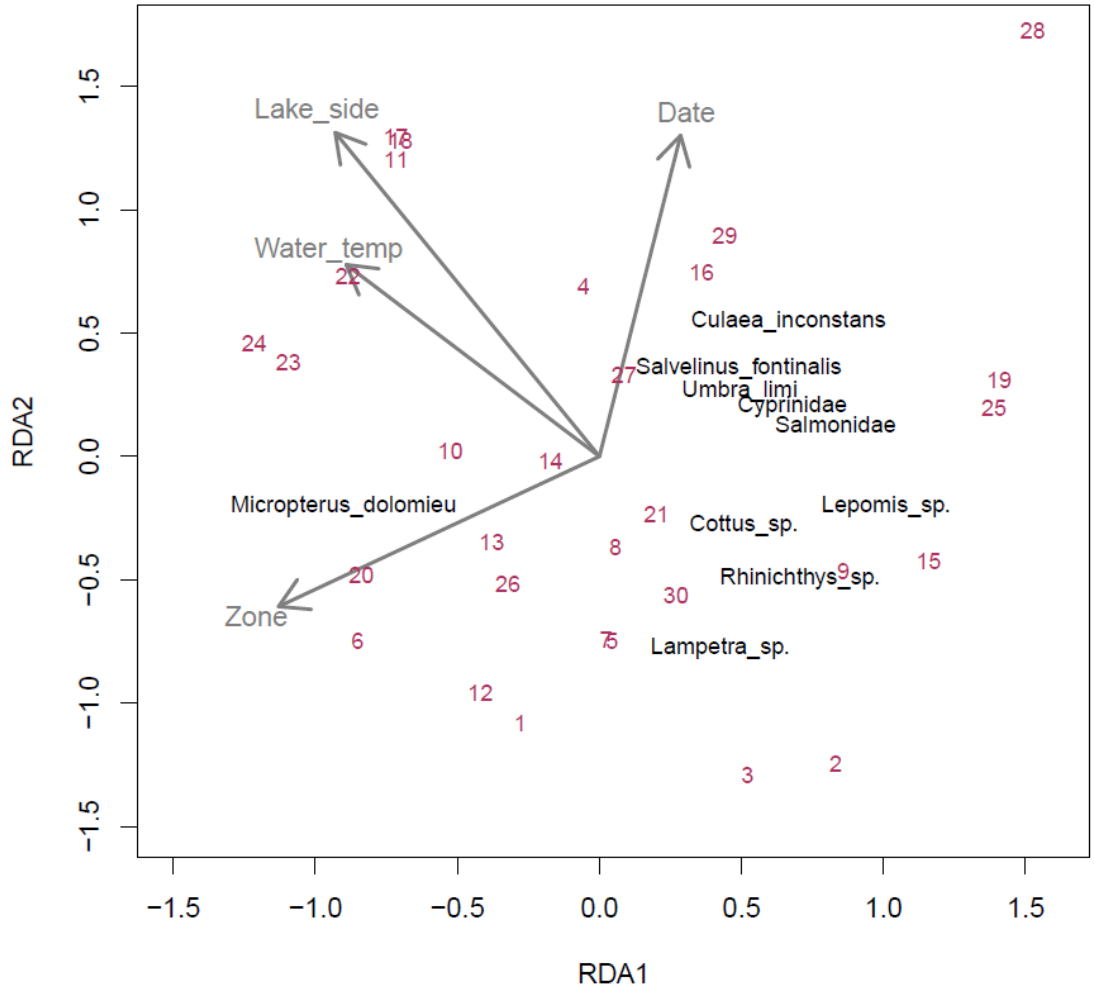


Figure 8: Redundancy analysis (RDA) plot demonstrating how four different variables represented with grey arrows (sampling date, sampling zone, lake side, and water temperature) influenced fish community composition in Boardman Lake measured with eDNA. Maroon numbers are net groups, and only the 10 taxa with the highest scores (black text) were included in the plot to facilitate visualization. All four variables were significant (p<0.05).



SUPPLEMENTARY MATERIALS

Supplementary tables and figures can be found in the Chapter 2 folder of the following shared Google Drive which can be accessed using the link below:

https://drive.google.com/drive/folders/1eRaQilvMqp6elFMC-EzZ7fL43_s-ghoU?usp=sharing

Table S1: Sampling metadata for all eDNA water samples collected in Boardman Lake and below Union Street Dam in the Boardman River.

Table S2: Number of unique taxa in each net group for each sampling method, which included eDNA (12S and 16S genes) and gill and fyke net sampling, with mean, standard deviation, and standard error. A net group consisted of one gill net and one fyke net with their corresponding water samples (3 replicates per net set).

Table S3: Counts and percentages of catches for all species caught in fyke nets and gill nets in Boardman Lake.

Figure S1: Boxplot of sequence read counts for 12S and 16S per water sample from Boardman Lake and below-dam Boardman River. The average number of reads in 12S was significantly higher than that of 16S ($p<0.001$).

Figure S2: Heatmap illustrating the number of replicates in which a taxon was present in each net group for 12S. A net group in Boardman Lake consisted of one gill net and one fyke net, each with 3 corresponding water samples, therefore the maximum number of detections for net groups 1-30 (Boardman Lake) is 6. Net groups 31 and 32 (below Union Street Dam in the Boardman River) consisted of 9 water samples and had no corresponding net sets, therefore anything between 6 and 9 detections is labeled (>6) for these two "net groups".

Figure S3: Heatmap demonstrating the number of replicates in which a taxon was present in each net group for 16S. A net group in Boardman Lake consisted of one gill net and one fyke net, each with 3 corresponding water samples, therefore the maximum number of detections for net groups 1-30 (Boardman Lake) is 6. Net groups 31 and 32 (below Union Street Dam in the Boardman River) consisted of 9 water samples and had no corresponding net sets, therefore anything between 6 and 9 detections is labeled (>6) for these two "net groups".

Figure S4: Correlation between the number of unique taxa detected and the number of reads for (a) 12S and (b) 16S for each eDNA water sample replicate from Boardman Lake and Boardman River. Spearman's correlation coefficient r_s was = 0.47 and 0.41 for 12S and 16S, respectively, and $p<0.001$ for both genes.

Figure S5: Correlation between the number of instances a taxon was detected using 12S and 16S in eDNA samples. Points represent all detected taxa and data include both Boardman Lake and below-dam Boardman River water samples. Pearson's correlation coefficient r was 0.91 and $p<0.001$.

Figure S6: Number of fish DNA reads in negative field controls associated with water samples from Boardman Lake and below Union St Dam in the Boardman River for 12S and 16S genes. Three large outliers were removed prior to analysis. The number of reads in negative controls for each gene did not significantly differ ($p=0.362$).

Figure S7: Correlation of species contamination in negative field controls from Boardman Lake and below Union St Dam in the Boardman River detected with 12S and 16S genes. Each point represents a fish species detected in negative controls. Three large outliers were removed prior to analysis. Pearson's correlation coefficient r was 0.75 and $p<0.001$.

Figure S8: Boxplot of average number of fish caught per net for fyke and gill nets in Boardman Lake. Gill nets caught significantly more fish on average than fyke nets ($p=0.004$).

Figure S9: Heatmap indicating taxa detected in each net group using traditional net sampling in Boardman Lake. Red indicates instances in which a species was only observed in a gill net, blue indicates species that were only observed in a fyke net, purple indicates when a species was detected in both net types, and grey represents no observation of that species in either net.

Figure S10: Pearson correlations between the number of fish of a particular species in a net and the number of DNA reads for that species in corresponding eDNA water samples from Boardman Lake. Correlations were performed only for the five most common species (*Ambloplites rupestris*, *Catostomus commersonii*, *Esox lucius*, *Sander vitreus*, and *Perca flavescens*) for each net type and gene. Each point represents an individual water sample. All correlations were insignificant except for *Perca flavescens* in fyke nets with 12S ($p=0.01$).

Figure S11: Pearson correlations between the biomass (g) of a particular species in a net and the number of DNA reads for that species in corresponding eDNA water samples from Boardman Lake. Correlations were performed only for the five most common species (*Ambloplites rupestris*, *Catostomus commersonii*, *Esox lucius*, *Sander vitreus*, and *Perca flavescens*) for each net type and gene. Each point represents an individual water sample. No correlations were significant at $\alpha=0.05$.

Figure S12: Plots of all eight non-metric multidimensional (NMDS) comparisons performed. Each point represents a net group. All taxa that were significant ($p<0.05$) in each analysis were included. 95% confidence interval ellipses were drawn around groups for all comparisons except 12e due to the downstream group only having two data points.

Figure S13: Redundancy analysis (RDA) plot demonstrating how four different variables represented with grey arrows (sampling date, sampling zone, lake side, and water temperature) influenced fish community composition in Boardman Lake estimated with traditional net sampling. Maroon numbers are net groups, and only the 10 taxa with the highest scores were included in the plot. No variables were significant in this analysis (all with $p>0.05$).

Supplementary File 1. Table with number of sequence reads for each taxa in each water sample replicate from Boardman Lake and Boardman River for the 16S gene, including field negative controls, lab negative controls, and lab positive controls. Metadata for each sample is included. Note that contamination in controls has not been subtracted from sample read counts listed here. See Supplementary file 3 for read counts in each sample after accounting for contamination in controls.

Supplementary File 2. Table with number of sequence reads for each taxa in each water sample replicate from Boardman Lake and Boardman River for the 12S gene, including field negative controls, lab negative controls, and lab positive controls. Metadata for each sample is included. Note that contamination in controls has not been subtracted from sample read counts listed here. See Supplementary file 3 for read counts in each sample after accounting for contamination in controls.

Supplementary File 3. Table with number of sequence reads for each taxa in each water sample replicate from Boardman Lake and Boardman River for the 16S gene after accounting for contamination in field and lab controls. Metadata for each sample is included.

Supplementary File 4. Table with number of sequence reads for each taxa in each water sample replicate from Boardman Lake and Boardman River for the 12S gene after accounting for contamination in field and lab controls. Metadata for each sample is included.